INVESTING & MARKETS (INDIA)

Trading: Strategy, Risk & Psychology

If you are going to trade, do it properly — the honest base rates, where an edge can and cannot come from, styles and timeframes costed out, writing a strategy as a specification, expectancy and R-multiples, backtesting without lying to yourself, position sizing and worked risk-of-ruin tables, stops and gap risk in Indian stocks, drawdown maths and a response protocol, costs and business-income tax, a full journal template, the psychology that actually decides outcomes, retail algo rules in India, and how the tip-service industry works.

trading · risk management · position sizing · expectancy · backtesting · psychology · drawdown · stop loss · india · nse · sebi · algo trading · trading journal · r-multiple · kelly criterion · costs

Study Pack · Comprehensive reference & interview prep · ~116 min read · 25,476 words

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Read this first: the base rates for retail trading

This guide takes trading seriously as a discipline. That means starting with the outcome data rather than the techniques, because any honest treatment of trading has to begin by establishing how unlikely success is and what distinguishes the small minority who achieve it.

This is financial education, not financial advice

Nothing here is a recommendation to buy, sell or trade any security, instrument or strategy. Active trading is high-risk and the evidence is that most individuals who attempt it lose money. Every rate, tax treatment, charge and regulation below is illustrative and dated **as at late 2026** — Indian market rules and tax provisions have changed repeatedly. Verify current details at **nseindia.com**, **sebi.gov.in** and **incometax.gov.in**. For personal advice, consult a SEBI-registered Investment Adviser (RIA). No strategy in this guide is presented as a way to make money.

What the Indian evidence shows

Source	Finding (as reported — verify the current version)
SEBI study on individual traders in the equity F&O segment, published September 2024, covering FY 2021–22 to FY 2023–24	Roughly **93% of individual F&O traders made losses**. Aggregate individual net losses across the three years were reported in the region of **₹1.8 lakh crore** (₹1,80,000 crore). Transaction costs were a material component of the loss.
SEBI follow-up study published in 2025, covering FY 2024–25	Roughly **91% of individual traders in index options** lost money, despite the regulatory curbs introduced from late 2024.
SEBI study on intraday equity (cash segment) trading	A large majority of individual intraday traders in the cash segment also made losses, with the loss rate increasing with the number of trades.
International academic literature on day trading	Studies of day traders in several markets have consistently found that a small minority achieve persistent profitability, that past profitability has limited predictive power, and that trading intensity correlates negatively with returns.

⚠ **Treat the numbers as directional, not immutable**
SEBI revises these studies, the definitions and the periods. Read the current primary source at **sebi.gov.in**. What is stable across every version: a large majority of individuals lose, losses are concentrated in small and high-frequency accounts, and transaction costs account for a meaningful share of the shortfall.

~93%
INDIVIDUAL F&O TRADERS LOSING, FY22–FY24

~91%
INDEX OPTION TRADERS LOSING, FY25

₹1.8 lakh cr
AGGREGATE INDIVIDUAL NET LOSS, FY22–FY24

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PROFITABLE SYSTEMS SOLD IN THIS GUIDE

Why this guide exists anyway

Given that evidence, there are two defensible reasons to read a serious guide on trading.

YOU ARE GOING TO TRY REGARDLESS

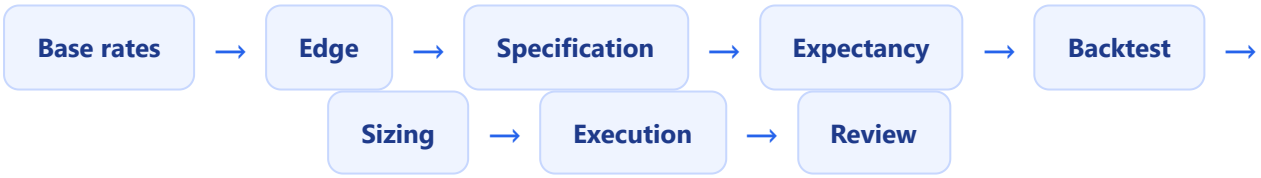
Many people will trade whatever the evidence says. If you are one of them, the difference between doing it with a written specification, measured expectancy, controlled position sizing and a journal, versus doing it on intuition and Telegram tips, is the difference between a survivable education and a destroyed balance sheet. This guide is the former.

THE DISCIPLINE TRANSFERS

Expectancy, risk of ruin, overfitting, out-of-sample validation, cost modelling and base rates are not trading concepts — they are decision-making concepts. An engineer who internalises them makes better decisions about system design, estimation, hiring and product bets. The maths here is worth knowing even if you never place a trade.

What this guide will not do

It will not give you a strategy, a setup, an indicator combination or a "system". Those are worthless when handed over, because any edge that can be written down and distributed to a million people stops being an edge. what it gives you is the *process* by which an edge would be specified, tested, sized, executed and reviewed — and the honest standards against which to judge whether you have one.



Trading versus investing: different games, different maths

These two words are used interchangeably in casual conversation and they describe activities with almost nothing in common. Confusing them is the first and most expensive error, because the strategies, the risk management, the tax treatment and the base rates are all different.

The structural comparison

Dimension	Investing	Trading
Source of return	Business earnings, reinvestment, economic growth — value created outside the market	Price movement, captured from another participant — value transferred within the market
Game type	Positive sum — all long-term holders can gain simultaneously	Approximately zero sum before costs; negative sum after
Holding period	Years to decades	Seconds to months
Decisions per year	A handful, often zero	Dozens to thousands
Cost drag	Low — perhaps 0.2–1.0% a year in an index fund	High — can exceed 30–40% a year at retail sizes and frequencies
Where skill is required	Mostly in behaviour: saving consistently, not panicking, not timing	In analysis, execution, risk control *and* behaviour, continuously
Time commitment	Hours per year	Hours per day
Scalability	Excellent — works identically at ₹10,000 and ₹10 crore	Poor — capacity constraints, impact cost, and fixed costs punish small accounts
Base rate of success	High for a diversified long-horizon index investor who stays invested	Low — the SEBI data above
Tax treatment (India)	Capital gains; long-term rates after the holding period	Generally business income (speculative for intraday equity; non-speculative for F&O), taxed at slab rates
What failure looks like	Underperformance over a decade	Capital destruction over months

✦ **A clarifying test**

Ask: "if the market closed for five years and reopened, would my position still make sense?" If yes, you are investing — you own a claim on earnings. If the question is absurd because your position depends entirely on price action next week, you are trading. Many people believe they are investing while behaving entirely as traders, which is how a "long-term holding" gets sold after a 12% drawdown.

The arithmetic difference, made concrete

TEXT

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TWO PEOPLE, Rs 5,00,000, TEN YEARS
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PERSON A - SYSTEMATIC INVESTOR
  Puts Rs 5,00,000 into a broad low-cost index fund.
  Adds Rs 20,000 per month.
  Assumed gross return 11% p.a. (illustrative - NOT a forecast;
  past performance does not predict future returns).
  Expense ratio 0.20%. Net 10.8%.
  Trades: 120 purchases over 10 years. Costs: negligible.
  Time spent: about 2 hours a year.

  Approximate terminal value:
    Lump sum: 5,00,000 x (1.108)^10        = Rs 13,93,000
    SIP: 20,000/month for 120 months
         at 0.9% monthly  (approx)          = Rs 43,70,000
                                             ---------------
    TOTAL (approx)                          = Rs 57,63,000
  Total contributed: 5,00,000 + 24,00,000   = Rs 29,00,000

PERSON B - ACTIVE TRADER
  Same Rs 5,00,000 and the same Rs 20,000 per month added.
  Trades 15 round trips per month.
  Average cost per round trip Rs 180 (brokerage, STT,
  exchange charges, GST, stamp duty, spread).
  Annual cost = 15 x 12 x 180 = Rs 32,400

  For Person B to MATCH Person A, B must generate 10.8% net
  PLUS Rs 32,400 a year of costs on a growing base.

  Year 1: capital about Rs 7,40,000 (5L + 12 x 20k)
          cost drag 32,400 / 7,40,000 = 4.4%
          required gross return = 15.2%
  Year 5: capital about Rs 20,00,000
          cost drag = 1.6%
          required gross return = 12.4%
  Year 10: capital about Rs 45,00,000
          cost drag = 0.7%
          required gross return = 11.5%

  SO: Person B must beat the index by 0.7-4.4 percentage points
  a year, every year, for a decade, net of tax, while spending
  hundreds of hours a year doing it, in a game where ~90% of
  participants lose.

  AND the tax treatment is worse: trading gains are business
  income at slab rates (up to 30% plus surcharge and cess),
  while long-term capital gains on equity have a lower rate
  and an annual exemption.
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  Verify current rates at incometax.gov.in - as at FY 2026-27.

THE OPPORTUNITY COST IS NOT JUST THE MONEY
  Person B spent roughly 1,500-3,000 hours over the decade.
  At a software engineer's marginal value of time, those hours
  invested in skills, side income or career progression would
  very plausibly have been worth more than any trading profit.
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The honest hybrid

✓ **If you want to trade, ring-fence it**

The defensible structure is a core long-term portfolio that is automated, boring and untouchable, plus a small, separately accounted trading allocation that you can lose entirely without consequence. Separate broker accounts if possible. Never pledge the core to fund the satellite. Never move money from the core to the satellite after a loss. Review the satellite's performance against the core annually and be prepared to shut it down.

Where an edge can come from — and where it cannot

An edge is a repeatable reason why your expected value is positive. There are a limited number of places an edge can come from, and for a retail participant most of them are structurally inaccessible. Knowing which is which saves years.

The six sources of edge

Source	What it means	Who has it	Realistic for retail in India?
Information	Knowing something material before the market prices it	Insiders (illegally), company management, large institutional research operations with channel checks and expert networks	**No** — and acting on genuine non-public price-sensitive information is insider trading, prohibited under SEBI regulations with serious penalties
Analysis	Interpreting public information better than the consensus	Skilled fundamental analysts, specialists in neglected small caps, people with genuine domain expertise	**Marginally** — possible in under-covered segments, over long horizons, with real work. Not possible in liquid large caps on a weekly timeframe.
Speed	Reacting to information or order flow faster than others	HFT firms with co-located servers, microwave links and custom hardware	**No** — the relevant timescale is microseconds. You are not in this race and never will be.
Structure	Capturing returns available because of market plumbing: market making, arbitrage, index rebalancing, borrow/lend spreads	Proprietary desks, market makers, large arbitrage books	**Largely no** — requires capital, infrastructure, negotiated costs and exchange memberships
Behaviour	Being systematically more disciplined than the average participant: not panicking, not chasing, following a plan	Anyone willing to do the work	**Yes** — this is the one genuinely accessible source, and it is also the hardest to maintain
Risk management	Surviving long enough for a small edge to	Anyone willing to do the work	**Yes** — and it is not optional. It is the only

	compound; sizing so no single outcome is fatal		thing standing between a small edge and ruin.

⚠ **The uncomfortable implication**

Four of the six sources of edge are closed to you. The two that remain — behaviour and risk management — are *defensive*. They stop you losing faster than necessary; they do not by themselves generate positive expectancy. A retail participant's realistic best case is a small analytical edge in a neglected corner, protected by excellent risk management and disciplined behaviour. Anyone offering you an edge in information, speed or structure is selling something.

Things frequently mistaken for an edge

Claimed edge	Why it is not one
"I understand technical analysis"	So does everyone with a charting app. Patterns visible to millions are priced in. The academic evidence on the profitability of common technical rules after costs is weak and regime-dependent. Some order-flow and volatility signals have content; "head and shoulders" does not.
"I read the news faster"	Algorithms parse news feeds in milliseconds. By the time you have read a headline on an app, the price has moved.
"I use a combination of five indicators"	All common indicators are deterministic functions of past price and volume. Combining them adds parameters, not information, and dramatically increases overfitting risk.
"I have a 90% accurate system"	Accuracy without loss magnitude is meaningless. Selling far out-of-the-money options gives a 90%+ win rate with negative expectancy. Ask for the maximum loss and the drawdown, not the win rate.
"I follow a guru with a great track record"	Unaudited track records, survivorship bias in who you hear about, and the fact that the guru's income comes from subscriptions rather than trading. See the scams section.
"I know the sector because I work in it"	Genuine domain knowledge is a real starting point for *analysis*, but only if applied over long horizons to under-covered companies. It is also the fastest route to inadvertently trading on material non-public information. Be extremely careful.
"I am good at maths / I am an engineer"	Helpful for building and testing systems; irrelevant to whether an edge exists. Many quantitatively excellent people have lost money because the edge was never there.
"It worked for the last three months"	Three months is noise. With a 55% win rate, a run of 20 wins in 30 trades has a probability of several percent — common enough that someone, somewhere, is always showing you one.

The edge question, asked properly

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THE FOUR QUESTIONS THAT TEST A CLAIMED EDGE
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1. WHO IS ON THE OTHER SIDE, AND WHY ARE THEY WRONG?
   Every trade has a counterparty. If you are buying, someone
   is selling. What do you know or do that they do not?
   Acceptable answers:
     "A forced seller (index rebalance, redemption, margin
      call) is transacting for non-informational reasons."
     "I am providing liquidity when it is scarce and being
      paid the spread for it."
     "I have analysed a neglected company more thoroughly
      than the two analysts who cover it."
   Unacceptable answers:
     "The chart looks bullish."
     "The indicator crossed over."
     "It has been going up."

2. WHY HAS THIS NOT BEEN ARBITRAGED AWAY?
   If an effect is real, persistent and easy to find, capital
   floods in and it disappears. Acceptable reasons for
   persistence:
     - Capacity constraints (the opportunity is too small for
       institutional money to bother with)
     - Structural barriers (regulation, mandates, taxation)
     - Behavioural causes that are deeply human and stable
     - It is compensation for a real risk you are bearing
   If you cannot name one, the effect is probably not real.

3. WHAT IS THE SAMPLE SIZE, AND WHAT IS THE OUT-OF-SAMPLE
   PERFORMANCE?
   Fewer than 100 trades is anecdote. In-sample performance
   is a measure of your curve-fitting ability, not of edge.

4. WHAT IS THE EXPECTANCY AFTER COSTS, AND WHAT IS THE
   MAXIMUM DRAWDOWN?
   If you cannot state both numbers, you have not tested it.

If your claimed edge fails any of these four, you do not have
an edge. You have a hypothesis, which is a fine thing to have
and a terrible thing to bet money on.
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Trading styles and timeframes

"Trading" covers activities as different from each other as sprinting is from marathon running. Each has distinct capital requirements, time demands, cost sensitivity and

difficulty. Choosing a style that is incompatible with your actual life is the most common structural mistake.

The styles, costed out

Style	Hold period	Trades/month	Realistic minimum capital	Time per day	C
Scalping	Seconds to minutes	200–1,000+	₹10 lakh+ (costs dominate below this)	Full session, uninterrupted	**E**
Intraday	Minutes to hours; flat by close	20–100	₹5 lakh+	4–6 hours	V
BTST / STBT	Overnight only	10–30	₹3 lakh+	1–2 hours	H
Swing	2–15 days	5–20	₹2 lakh+	30–60 min	M
Positional	Weeks to months	2–8	₹2 lakh+	30 min, a few times a week	L
Pairs / statistical arbitrage	Days to weeks	Varies	₹10 lakh+ (two legs, margin)	Depends on automation	H

The strategy archetypes

MOMENTUM / TREND FOLLOWING

Buy what is rising, sell what is falling. Premise: price trends persist over certain horizons. Characteristics: low win rate (30–45%), large average winner, many small losers, long flat periods, severe drawdowns when trends reverse. Requires the discipline to take many small losses and let a few winners run — psychologically the hardest profile.

MEAN REVERSION

Buy what has fallen too far, sell what has risen too far. Premise: prices overshoot and revert. Characteristics: high win rate (60–75%), small average winner, occasional very large loser when "too far" keeps going. Psychologically comfortable and structurally dangerous — it is the same payoff shape as selling options.

BREAKOUT

Enter when price exits a defined range. Premise: consolidation precedes directional expansion. Characteristics: many false breakouts, cost-heavy, highly sensitive to the definition of the range and to slippage at the moment of the break, when liquidity is thinnest.

PAIRS / RELATIVE VALUE

Long one instrument, short a related one; profit from the spread converging. Premise: the relationship is stable. Characteristics: market-neutral-ish, two sets of costs, requires borrow or derivatives for the short leg, and fails catastrophically when the relationship breaks for a fundamental reason.

EVENT DRIVEN

Position around results, corporate actions, index inclusions, policy decisions. Premise: predictable flows or predictable reactions. Characteristics: binary outcomes, high implied volatility, you are competing with specialists, and the event is already priced.

VOLATILITY

Trade implied versus realised volatility rather than direction. Premise: options are systematically mispriced in one direction. Characteristics: requires option fluency and Greeks management; for retail it almost always degenerates into naked option selling, which is covered as a hazard in the derivatives guide.

Win rate and payoff ratio are not independent

Across almost all strategy archetypes, the two trade off against each other. Momentum gives you a low win rate and a high payoff ratio. Mean reversion gives you a high win rate and a low payoff ratio. There is no archetype offering both, because if there were, it would be arbitraged. When someone advertises "75% accuracy with 1:3 risk-reward", they are describing a combination that would be extraordinarily profitable and is therefore almost certainly a curve-fitted backtest or a fabrication.

Choosing a style: the honest matching exercise

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STYLE SELECTION WORKSHEET
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Answer these before choosing anything.

1. HOW MANY UNINTERRUPTED HOURS CAN YOU GIVE TO THE MARKET
   DURING SESSION HOURS, EVERY DAY, SUSTAINABLY?
     0 hours          → positional only, or do not trade
     under 1 hour     → positional
     1–2 hours        → swing or positional
     4+ hours, and
     your employer is
     fine with it     → intraday is arithmetically possible
     (If you are employed full-time as a software engineer,
      the honest answer is 0–1. Act accordingly.)

2. WHAT IS YOUR RISK CAPITAL – money whose total loss changes
   nothing material in your life?
     under Rs 2 lakh  → do not trade; costs will dominate
     Rs 2–5 lakh      → positional/swing, small size
     Rs 5 lakh+       → more styles become arithmetically
                        viable (not more likely to succeed)

3. CAN YOU TOLERATE A 30% DRAWDOWN WITHOUT CHANGING YOUR
   BEHAVIOUR?
     No  → trend following and momentum are out. Your
           drawdowns will be deeper than you expect.
     Yes → prove it. Most people say yes and discover
           otherwise at −12%.

4. DO YOU PREFER BEING RIGHT OFTEN, OR MAKING MORE WHEN RIGHT?
     Right often      → mean reversion. BEWARE: this is the
                        profile that hides a fat left tail.
     More when right  → momentum. You will be wrong 60% of
                        the time and must be fine with that.

5. CAN YOU WRITE CODE AND MANAGE DATA?
     Yes → systematic testing is available to you, which is
           a genuine advantage over discretionary traders.
     No  → you will be relying on discretion, which cannot
           be backtested, which means you cannot know whether
           you have an edge. This is a serious limitation.

THE MOST COMMON MISMATCH
  A full-time employed engineer choosing intraday or expiry-day
  trading. It fails on question 1, every time, and the failure
  is expensive because the style with the least compatible time
  profile is also the one with the highest cost sensitivity.
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Market structure for traders

Before any strategy, you need to understand the mechanics of how your order becomes a trade and what it costs you in ways that never appear on a contract note. Market structure is the terrain; strategy is the route.

Liquidity and the bid–ask spread

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READING AN ORDER BOOK
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A hypothetical liquid large-cap share:

   BIDS (buyers)              ASKS (sellers)
   Qty      Price             Price    Qty
   ─────    ───────           ───────  ─────
   4,200    1,799.80          1,800.00  3,800
   6,500    1,799.70          1,800.10  5,100
   9,100    1,799.60          1,800.20  7,400
  12,000    1,799.50          1,800.30 10,900
  15,500    1,799.40          1,800.40 14,200

  BEST BID  = 1,799.80   BEST ASK = 1,800.00
  SPREAD    = 0.20       = 0.011% of price
  MID       = 1,799.90

WHAT A MARKET ORDER COSTS YOU
  Buy 3,000 shares at market → filled at 1,800.00 (enough
  depth at the best ask). Cost vs mid = 0.10 x 3,000 = Rs 300.

  Buy 20,000 shares at market → you consume multiple levels:
      3,800 @ 1,800.00 = 68,40,000
      5,100 @ 1,800.10 = 91,80,510
      7,400 @ 1,800.20 = 1,33,21,480
      3,700 @ 1,800.30 = 66,61,110
      ──────────────────────────────
     20,000 shares, avg price 1,800.15
  Cost vs mid = 0.25 x 20,000 = Rs 5,000
  This extra 0.15 is IMPACT COST - you moved the price
  against yourself by trading.

NOW THE SAME EXERCISE ON A THIN STOCK
   BIDS                       ASKS
   Qty     Price              Price   Qty
   ─────   ──────             ──────  ─────
     200   412.05             413.90    150
     450   411.60             414.70    380
     800   410.20             416.00    600

  SPREAD = 1.85 = 0.45% of price
  Buy 1,000 shares at market:
      150 @ 413.90, 380 @ 414.70, 470 @ 416.00
      avg = 415.26 vs mid of 412.98
  IMPACT = 0.55% on entry. Another 0.55%+ on exit.
  ROUND-TRIP STRUCTURAL COST: over 1.1%, before brokerage.

  A strategy with a 0.8% average gross edge is DEAD in this
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stock and ALIVE in the liquid one. Same strategy. The
instrument decides.
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Liquidity measure	What it tells you	Rule of thumb
Bid–ask spread (%)	Immediate round-trip friction	Your average expected gross profit should be at least 5–10x the spread
Depth at best bid/ask	How much you can trade without moving the price	Your order should be a small fraction of displayed depth
Average daily traded value	Overall capacity	Your position should be well under 1% of ADTV; smaller for intraday
Impact cost (published by exchanges)	Estimated cost of executing a standard order size	NSE publishes impact cost for index constituents; check it before assuming liquidity
Spread stability through the day	Whether liquidity disappears when you need it	Spreads widen at the open, at the close, and during news. Those are exactly the times you want to trade.

⚠ **Thin stocks punish you twice**

First on entry and exit through the spread and impact. Second, and worse, when you need to exit urgently — a stock that trades ₹2 crore a day in normal conditions may have no bid at all on a bad day. Circuit limits can freeze it entirely. Every backtest that includes illiquid names and assumes you got the closing price is lying to you, and the lie is largest precisely in the trades that matter most.

Circuit limits and gap risk in Indian equities

Mechanism	What it does	Consequence for a trader
Stock-level price bands	Individual securities have daily movement limits (commonly 2%, 5%, 10% or 20%, depending on the security; securities in the F&O segment have different treatment, with dynamic price bands)	If a stock hits the upper or lower circuit, you may be **unable to exit at any price**. A lower-circuit stock can be locked for consecutive sessions.
Index-level market-wide circuit breakers	Trading halts across the market at defined index moves (commonly cited trigger levels are 10%, 15% and 20%), with halt durations depending on the level and the time of day	Trading stops entirely. Your stop losses do not execute. Your margin calls do not wait.
Dynamic price bands / operating ranges	Intraday bands that flex, used in F&O-eligible securities and derivatives to curb erroneous orders	Orders outside the band are rejected; the band may be relaxed by the exchange
Overnight gaps	Price discontinuity between one close and the next open, caused by news outside market hours	**No stop loss protects against a gap.** This is the single most under-appreciated risk for positional and BTST traders.

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THE GAP MATHS - WHY A STOP LOSS IS NOT A GUARANTEE
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You are long 1,000 shares at Rs 1,800 = Rs 18,00,000
Stop loss placed at Rs 1,710 (-5%)
Planned maximum loss = Rs 90,000

SCENARIO A - ORDERLY DECLINE
  Price drifts down intraday, your stop triggers at 1,710,
  you are filled at 1,709.50.
  Actual loss = Rs 90,500. The plan worked.

SCENARIO B - OVERNIGHT NEWS, GAP DOWN
  Previous close 1,790. Overnight: a regulatory action is
  announced. Stock opens at 1,520 (-15%).
  Your stop at 1,710 becomes a MARKET ORDER at the open.
  Filled at roughly 1,518.
  Actual loss = (1,800 - 1,518) x 1,000 = Rs 2,82,000

  THREE TIMES your planned maximum loss. The stop did exactly
  what it was designed to do and was irrelevant.

SCENARIO C - GAP DOWN TO LOWER CIRCUIT
  Stock opens at the 20% lower circuit, 1,432, locked.
  No buyers. Your stop cannot execute.
  Day 2: opens at lower circuit again, 1,146.
  Day 3: opens at lower circuit again, 917.
  You finally exit at 920.
  Actual loss = (1,800 - 920) x 1,000 = Rs 8,80,000
  Nearly TEN TIMES your planned maximum loss.

  This is not hypothetical. Indian mid- and small-cap stocks
  have experienced multi-day lower-circuit sequences following
  governance events, regulatory actions and fraud disclosures.

THE IMPLICATIONS FOR POSITION SIZING
  Your "maximum loss" from a stop is a planning figure, not a
  guarantee. Size positions so that a 3x-worse-than-planned
  outcome is survivable, not fatal.

  If your stop implies a 2% account loss, ask: "can I survive
  a 6% loss on this position?" If no, the position is too big.

  This is why single-stock concentration is so much more
  dangerous than index exposure: an index cannot be locked at
  the lower circuit for three days because of one company's
  fraud, but a stock can.
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Circuit mechanisms and band levels illustrative. As at late
2026 – verify current rules at nseindia.com and sebi.gov.in.
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Order types and what they actually do

Order type	Behaviour	When to use	Hazard
Market	Executes immediately at the best available price(s)	Only in deeply liquid instruments when immediacy matters more than price	In a thin book you can be filled far from the last price. Never use in options.
Limit	Executes only at your price or better	Default for almost everything	May not execute at all — which is sometimes valuable information
Stop loss (SL-M)	Becomes a market order when the trigger is hit	When you must exit	Fills at whatever is available; useless against a gap
Stop loss limit (SL-L)	Becomes a limit order when the trigger is hit	When you want price control on the exit	May not fill at all in a fast move, leaving you in a losing position with no protection
Cover / bracket orders	Broker-level constructs bundling entry, stop and target	Intraday convenience	Availability has changed following regulatory revisions; verify with your broker. Do not rely on a broker product as your risk framework.
AMO (after-market order)	Queued for the next session's open	Placing orders outside hours	Executes into the opening auction, often the worst price of the day
GTT (good till triggered)	Broker-held conditional order valid for a long period	Longer-horizon alerts and exits	Broker-side, not exchange-side; depends on the broker's systems being up

✦ **The opening and closing minutes**

Spreads are widest, volatility highest and information lowest in the first and last few minutes of the session. A large proportion of retail intraday damage happens in those windows. If you must trade, waiting 15–30 minutes after the open costs you nothing and removes a category of avoidable loss.

Building a strategy: writing it as a specification

A strategy that is not written down is not a strategy — it is a mood. As a software engineer you already know the discipline: a system you cannot specify is a system you cannot test, and a system you cannot test is a system you cannot trust. Trading is the same, with money attached.

The seven components

1. **Hypothesis**
   A falsifiable statement about market behaviour, with a reason why it should be true and why it has not been arbitraged away. Written first, before any chart is opened.
2. **Universe**
   Exactly which instruments are eligible, defined by objective, checkable criteria — liquidity, price, market cap, segment membership. No discretion.
3. **Timeframe**
   The bar interval for signals, the expected holding period, and the review cadence.
4. **Entry rule**
   The precise, computable condition that triggers a trade. If two people applying it to the same data would get different answers, it is not a rule.
5. **Exit rule**
   The condition for taking profit. Separate from the stop.
6. **Stop rule**
   The condition for accepting the trade is wrong. Defined in advance, in price terms, before entry.
7. **Filter and sizing rules**
   Conditions under which you do not trade at all (regime filters), and how position size is determined from account equity and the stop distance.

> **The two-person test**
>
> Hand your written specification to a competent stranger with the same data. If they cannot reproduce your trade list exactly, your specification has discretion hidden in it, and that discretion is where your backtest results came from. Discretionary trading is legitimate, but it cannot be backtested, which means you can never know whether you have an edge — only whether you have had good luck.

A full worked strategy specification

TEXT

```
STRATEGY SPECIFICATION
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NAME        : Index Pullback Continuation (ILLUSTRATIVE ONLY)
VERSION     : 0.3
AUTHOR      : (you)
DATE        : (today)
STATUS      : UNTESTED HYPOTHESIS — NOT FOR LIVE CAPITAL

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1. HYPOTHESIS
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STATEMENT:
  In an established medium-term uptrend, short-term pullbacks
  to a defined support reference are followed, more often than
  chance, by a resumption of the trend over the following
  5–10 sessions.

WHY MIGHT THIS BE TRUE?
  Possible behavioural mechanism: participants who missed the
  initial move use pullbacks as entry points, creating a
  concentration of resting demand at reference levels.
  Possible structural mechanism: systematic trend-following
  allocators rebalance into weakness within a trend.

WHY MIGHT IT NOT BE ARBITRAGED AWAY?
  Honest answer: IT PROBABLY HAS BEEN. This is one of the most
  widely known patterns in existence. Any surviving edge is
  likely small, regime-dependent and smaller than transaction
  costs for most participants.
  THIS IS A RED FLAG AND IT IS WRITTEN DOWN DELIBERATELY.

FALSIFICATION CRITERION:
  If, over 200+ out-of-sample trades, expectancy after modelled
  costs is not greater than +0.10R, the hypothesis is rejected
  and the strategy is abandoned. No tweaking.

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2. UNIVERSE
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  - Instrument: a broad equity index, traded via index futures
    or via a liquid index ETF
  - Rationale for index over single stocks: no single-company
    fraud risk, no multi-day lower-circuit risk, tightest
    spreads, published impact cost
  - EXCLUDED: all single stocks (gap and circuit risk), all
    instruments with average daily traded value below a
    stated threshold
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3. TIMEFRAME
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  Signal bar        : daily close
  Decision time     : after the close; orders placed for the
                      next session
  Expected hold     : 5–10 sessions
  Maximum hold      : 15 sessions (time stop)
  Review cadence    : weekly log review; monthly metrics;
                      quarterly go/no-go

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4. ENTRY RULE  (all conditions must be true at the close)
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  E1. Close is above the 200-period simple moving average
      (regime: medium-term uptrend)
  E2. The 50-period SMA is above the 200-period SMA
  E3. Today's close is below the 20-period SMA
      (short-term pullback has occurred)
  E4. Today's close is above the 50-period SMA
      (the pullback has not broken the medium-term structure)
  E5. Today's close is greater than yesterday's close
      (first sign of resumption – avoids catching a knife)
  E6. India VIX is below 22 (regime filter – do not trade
      this pattern in a high-volatility regime)

  ORDER: buy at the next session's open.
  NO DISCRETION. If all six are true, the trade is taken.
  If any is false, no trade. There is no "but it looks good".

──────────────────────────────────────────────────────────────
5. EXIT RULE (profit)
──────────────────────────────────────────────────────────────

  X1. Close above the 20-period SMA by more than 0.5 x ATR(14)
      → exit at next open
  OR
  X2. Price reaches entry + 2.5 x ATR(14 at entry)
      → exit on touch
  OR
  X3. 15 sessions elapsed → exit at next open (time stop)

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6. STOP RULE
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  S1. Initial stop = entry – 1.5 x ATR(14 at entry)
  S2. Stop is never widened. Ever. Under any circumstances.
  S3. After price reaches entry + 1.5 x ATR, the stop moves
      to breakeven (entry price)
  S4. Hard stop: if the index closes below its 200-SMA, exit
```

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      all positions at the next open regardless of P&L
      (the regime that justified the trade has ended)

_____________________________________________________________

7. FILTERS AND SIZING
_____________________________________________________________

  F1. No new entries in the 3 sessions before a scheduled
      monetary policy decision or the Union Budget
  F2. Maximum 2 concurrent positions
  F3. Maximum portfolio heat 4% (sum of all open risk)
  F4. No trading for 3 sessions after 3 consecutive losses
      (circuit breaker - see the psychology section)

  SIZING:
    Risk per trade        = 1.0% of account equity
    Stop distance         = 1.5 x ATR(14) in points
    Position size (units) = (equity x 0.01) / stop distance
    Rounded DOWN to the nearest tradeable unit.
    If one unit exceeds the risk budget, NO TRADE.

_____________________________________________________________

8. COST MODEL (must be applied in every backtest)
_____________________________________________________________

  Brokerage           : Rs 20 per order + 18% GST
  STT                 : as applicable to the instrument
  Exchange charges    : as published
  SEBI turnover fee   : as published
  Stamp duty          : as applicable, buy side
  SLIPPAGE ASSUMPTION : 0.05% of notional per side, on top of
                        all of the above, on every fill,
                        including stop-loss fills
  GAP ASSUMPTION      : stop fills are modelled at the WORSE
                        of (stop price) and (next open)

_____________________________________________________________

9. METRICS TO RECORD
_____________________________________________________________

  Trades, win rate, average win (R), average loss (R),
  expectancy (R), profit factor, maximum drawdown (%),
  maximum drawdown duration (days), longest losing streak,
  exposure (% of time in market), return on risk capital,
  total costs as a % of gross P&L

_____________________________________________________________

10. KILL CRITERIA
_____________________________________________________________

  K1. Out-of-sample expectancy below +0.10R over 200 trades
  K2. Live drawdown exceeds 1.5x the worst backtested drawdown
  K3. Live expectancy below zero over 100 live trades
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  K4. Any rule has been changed more than twice since the
      last out-of-sample test (overfitting indicator)

  If a kill criterion triggers, the strategy stops. Permanently
  or until a full re-test on new data. NOT "with a small tweak".

THIS IS AN ILLUSTRATIVE SPECIFICATION FORMAT, NOT A
RECOMMENDED STRATEGY. The hypothesis is almost certainly not
profitable after costs. It is written out in full so you can
see what a complete specification looks like.
```

⚠ **Notice what section 1 does**

The hypothesis section openly states that the pattern is widely known and probably arbitraged away, and commits to a falsification criterion before any testing. That intellectual honesty at the start is what separates research from rationalisation. The overwhelming temptation in strategy development is to write the hypothesis *after* seeing the backtest, which converts a scientific process into a storytelling one.

Specification anti-patterns

Anti-pattern	What it looks like	Why it fails
Hidden discretion	"Enter on a breakout *if the setup looks clean*"	Untestable. Your backtest measures your hindsight, not the rule.
Undefined exit	"Hold until the trend ends"	Exits determine most of your return distribution. An undefined exit is an undefined strategy.
Stop as an afterthought	"I'll cut it if it goes too far"	"Too far" is defined by your emotional state at the moment of pain, which is the worst possible input.
No universe definition	"Any stock showing good momentum"	Survivorship bias and selection bias are guaranteed. You will test on the stocks you remember.
Parameter stacking	Eight indicators each with two tunable parameters	16 degrees of freedom on a few hundred trades. You are fitting noise with near-certainty.
No cost model	Backtest on closing prices with zero friction	Covered in detail in the backtesting section. This single omission invalidates most retail backtests.
No kill criteria	"I'll keep refining it"	Without a pre-committed stopping rule, you will refine until it looks good, which is the definition of overfitting.

Expectancy maths

Expectancy is the single number that determines whether a strategy makes money. Everything else — win rate, accuracy, profit factor, Sharpe ratio — is either derived from it or a distraction from it. If you learn one piece of arithmetic from this guide, learn this one.

R-multiples: the unit that makes everything comparable

TEXT

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THE R-MULTIPLE
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R = the amount of money you risked on a trade, defined as
    the distance from entry to stop, times the position size.

Every outcome is then expressed as a multiple of R:
  Lost at the stop        = -1.0R
  Exited at half the stop = -0.5R
  Made twice the risk     = +2.0R
  Gapped through the stop = -2.3R   (this happens)

WHY THIS IS THE RIGHT UNIT
  1. It normalises across position sizes and instruments.
     A Rs 4,000 loss on a small position and a Rs 40,000 loss
     on a large one are both -1.0R if both were sized to risk
     1% of equity.
  2. It separates the STRATEGY'S quality from the SIZING
     decision. A strategy with +0.3R expectancy is good
     whether you risk 0.5% or 2% per trade; the sizing
     decision only changes how fast you compound and how
     likely you are to be ruined.
  3. It makes expectancy directly computable.

WORKED EXAMPLE
  Account                Rs 5,00,000
  Risk per trade         1% = Rs 5,000  → 1R = Rs 5,000
  Entry                  25,000 (index units)
  Stop                   24,800  → stop distance 200 points
  Position size          5,000 / 200 = 25 units

  Exit at 25,340 → profit = 340 x 25 = Rs 8,500
  In R terms: 8,500 / 5,000 = +1.70R

  Exit at stop → loss = 200 x 25 = Rs 5,000 = -1.00R

  Gap down, filled at 24,610 → loss = 390 x 25 = Rs 9,750
  In R terms: -1.95R   ← RECORD THIS HONESTLY. Hiding gap
  losses by calling them -1R is how people convince themselves
  their risk control works.
```

The expectancy formula

TEXT

```
EXPECTANCY
==========
  E (in R) = (W x A_win) - (L x A_loss)

  W      = win rate (as a decimal)
  L      = loss rate = 1 - W
  A_win  = average winning trade, in R
  A_loss = average losing trade, in R (positive number)

EQUIVALENTLY, using the PAYOFF RATIO P = A_win / A_loss
  and assuming A_loss = 1.0R:

  E = (W x P) - (1 - W)

PROFITABILITY CONDITION:  E > 0  after costs.

WORKED - THREE STRATEGIES WITH IDENTICAL "QUALITY" CLAIMS
---------------------------------------------------------

STRATEGY A - "high accuracy mean reversion"
  W = 0.75,  A_win = 0.5R,  A_loss = 1.6R
  E = (0.75 x 0.5) - (0.25 x 1.6)
    = 0.375 - 0.400
    = -0.025R   LOSES MONEY despite a 75% win rate

STRATEGY B - "trend following"
  W = 0.35,  A_win = 3.2R,  A_loss = 1.0R
  E = (0.35 x 3.2) - (0.65 x 1.0)
    = 1.120 - 0.650
    = +0.470R   MAKES MONEY despite being wrong 65% of the time

STRATEGY C - "balanced swing"
  W = 0.52,  A_win = 1.4R,  A_loss = 1.0R
  E = (0.52 x 1.4) - (0.48 x 1.0)
    = 0.728 - 0.480
    = +0.248R

TRANSLATE INTO RUPEES
  With 1R = Rs 5,000 and 100 trades a year:
    A: 100 x -0.025 x 5,000 = -Rs   12,500
    B: 100 x +0.470 x 5,000 = +Rs 2,35,000
    C: 100 x +0.248 x 5,000 = +Rs 1,24,000

  Then SUBTRACT COSTS. At Rs 180 per round trip:
    100 trades x Rs 180 = Rs 18,000
    A: -Rs 30,500   B: +Rs 2,17,000   C: +Rs 1,06,000

THE LESSON THAT CANNOT BE OVERSTATED
  WIN RATE ALONE TELLS YOU NOTHING.
```

```
Strategy A wins three times as often as B and loses money.
Anyone who quotes accuracy without payoff ratio is either
confused or selling you something.
```

The win rate / payoff ratio frontier

TEXT

BREAK-EVEN WIN RATE FOR A GIVEN PAYOFF RATIO

Setting E = 0 and solving:

$$(W \times P) - (1 - W) = 0$$
$$W \times P + W = 1$$
$$W = 1 / (1 + P)$$

Payoff ratio P	Break-even win rate	Comment
0.25	80.0%	Selling tails. The structure of a short option book.
0.50	66.7%	Classic mean reversion
0.75	57.1%	
1.00	50.0%	A coin flip
1.50	40.0%	
2.00	33.3%	
2.50	28.6%	
3.00	25.0%	Classic trend following
4.00	20.0%	
5.00	16.7%	Long-tail momentum

NOW ADD COSTS. Suppose costs equal 0.15R per round trip (a realistic retail figure when 1R is small).

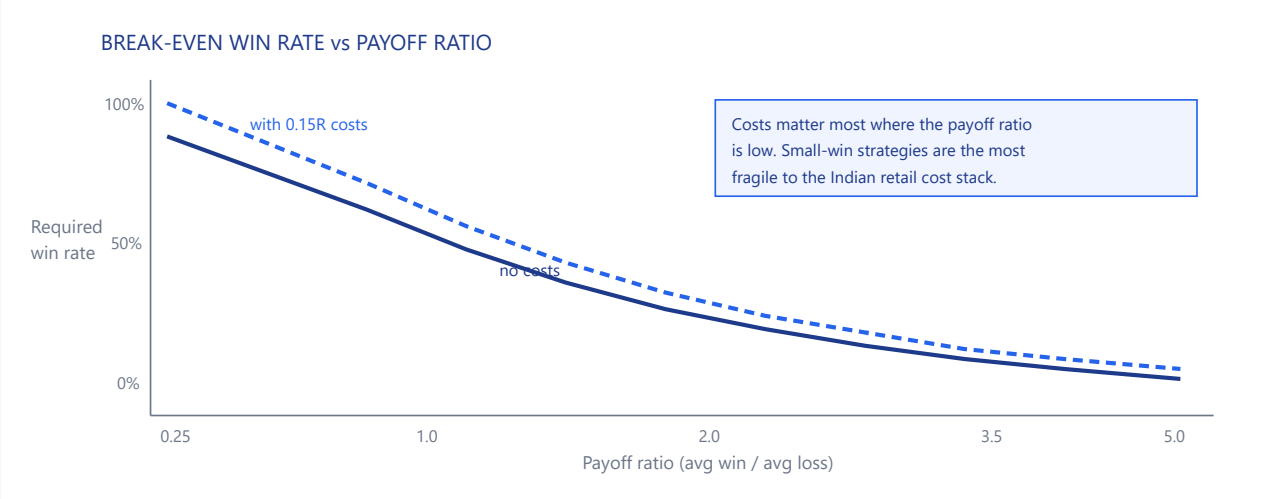
$$E = (W \times P) - (1 - W) - 0.15 = 0$$
$$W = 1.15 / (1 + P)$$

Payoff ratio P	BE win rate (no costs)	BE win rate (0.15R costs)	Extra win rate needed
0.25	80.0%	92.0%	+12.0 pp
0.50	66.7%	76.7%	+10.0 pp
1.00	50.0%	57.5%	+7.5 pp
2.00	33.3%	38.3%	+5.0 pp
3.00	25.0%	28.8%	+3.8 pp
5.00	16.7%	19.2%	+2.5 pp

READ THE LEFT-HAND SIDE OF THAT TABLE.
A strategy with a 0.25 payoff ratio needs a 92% win rate after costs. That is the arithmetic reason why "sell premium for small consistent gains" fails at retail cost levels. The costs eat a larger fraction of a small average win.

READ THE RIGHT-HAND SIDE.
High-payoff strategies are far more ROBUST to costs. If your average winner is 5R, a 0.15R cost is 3% of the winner. If your average winner is 0.25R, the same cost is 60% of it.

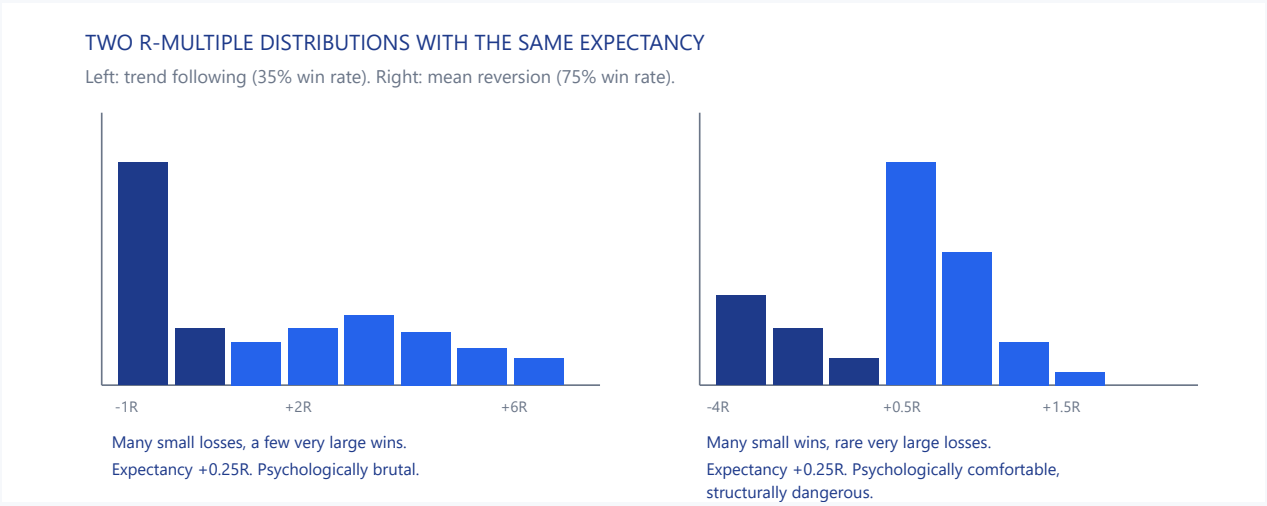
```
PRACTICAL CONSEQUENCE FOR A SMALL ACCOUNT
  Prefer strategies with a high payoff ratio and a low trade
  frequency. The cost structure of retail trading in India
  makes high-frequency, small-edge strategies arithmetically
  unavailable to you, regardless of how good the signal is.
```



The break-even frontier. Strategies to the left (small wins, high accuracy) require near-impossible win rates once realistic costs are included.

Profit factor, Sharpe and the metrics worth tracking

Metric	Formula	What it tells you	Caveat
Expectancy (R)	$(W \times A_{win}) - (L \times A_{loss})$	Expected profit per trade, in risk units	Needs 100+ trades to be meaningful
Expectancy (₹)	Expectancy(R) × 1R value	Expected rupees per trade	Only valid if 1R is consistent
Profit factor	Gross profit / gross loss	>1 means profitable; 1.5–2.0 is good; >3 on a small sample is a red flag	Extremely sensitive to one large winner
Win rate	Wins / total trades	Almost nothing on its own	The single most over-quoted and least informative metric
Maximum drawdown	Largest peak-to-trough equity decline	What you must survive emotionally and financially	Your future worst drawdown will likely exceed your backtested one
Drawdown duration	Time from peak to recovery of that peak	Often more psychologically destructive than depth	Routinely ignored; should not be
Sharpe-type ratio	(Return − risk-free) / volatility of returns	Return per unit of volatility	Penalises upside volatility; assumes normality; easily gamed by strategies with hidden tail risk
Exposure	% of time with capital at risk	Two strategies with the same return but different exposure are not equivalent	Low exposure with good returns is genuinely valuable
Cost ratio	Total costs / gross profit	How much of your edge the market takes	If this exceeds 40–50%, reduce frequency, not position size



Identical expectancy, opposite shapes. The left profile tests your patience; the right profile hides a fat left tail and tests your survival. Win rate alone cannot distinguish them.

⚠ **How many trades before expectancy means anything?**
With a true expectancy of +0.25R and a typical trade-to-trade standard deviation of around 1.5R, the standard error of your measured expectancy after *n* trades is roughly 1.5/√n. After 30 trades that is ±0.27R — wider than the effect you are trying to measure. After 100 trades it is ±0.15R. After 400 it is ±0.075R. **Fewer than 100 trades tells you essentially nothing.** Most retail traders form a strong conviction after 15.

Backtesting properly

A backtest is a simulation of a strategy against historical data. Done well, it is the only way to estimate expectancy before risking money. Done the way most retail traders do it, it is an elaborate machine for generating false confidence. The difference is entirely in the discipline, not in the tooling.

The data problems that come first

Problem	What it does to your results	Mitigation
Survivorship bias	If your universe is "current index constituents", you have excluded every company that was delisted, merged away or collapsed. Results are inflated, often dramatically.	Use point-in-time constituent lists. If you cannot obtain them, restrict to index-level instruments where the problem does not arise.
Corporate actions	Splits, bonuses, rights issues and demergers create apparent price crashes that your strategy will "trade".	Use properly adjusted series. Verify by checking known split dates.
Dividend adjustment	Price-only series understate total return; adjusted series alter historical levels.	Be explicit about which you are using and be consistent.
Bad ticks and stale prices	A single erroneous print can create a phantom profitable trade, especially in stop-based strategies.	Filter outliers; cross-check against a second source; inspect your largest winners individually.
Contract rollover (futures)	Naively concatenating contract months creates artificial gaps at each roll.	Use a continuous series with a documented roll and adjustment method; test sensitivity to the method.
Missing sessions	Indian market holidays, muhurat sessions and special sessions break naive date arithmetic.	Use an exchange trading calendar, not a generic business-day calendar.
Index methodology changes	Index composition rules and reconstitutions change over time.	Understand what the historical series actually represents before drawing conclusions from it.

Look-ahead bias: the error you will make

TEXT

```
LOOK-AHEAD BIAS - FOUR WAYS IT CREEPS IN
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1. USING THE CLOSE TO DECIDE AND THE CLOSE TO TRADE
   "If today's close is above the 20-SMA, buy at today's
   close." You did not know today's close until it happened.
   FIX: decide on close, execute at the NEXT open, and model
   the gap.

2. USING THE DAY'S HIGH OR LOW IN A SIGNAL
   "Buy if price touches the lower band and closes green."
   Within a single bar you cannot know the order of events.
   Did it touch the band before or after the close formed?
   FIX: use intraday data, or restrict signals to values
   known at bar close.

3. USING RESTATED OR LATE-ARRIVING DATA
   Fundamental data is published weeks after the period it
   describes, and is sometimes restated. Using a financial
   ratio "as at 31 March" on 1 April is look-ahead.
   FIX: lag fundamental data by the actual publication delay.

4. SURVIVORSHIP IN RULE DESIGN
   You chose the 20-period SMA because you have been watching
   this market for five years and 20 "works". That knowledge
   came from the data you are now testing on. This is the
   subtlest and most pervasive form, and it is unfixable by
   code. The only defence is genuinely held-out data.

THE DIAGNOSTIC THAT CATCHES MOST CASES
  If your backtest equity curve is remarkably smooth, or your
  win rate is above 70% with a payoff ratio above 1.5, or your
  maximum drawdown is under 8%, assume look-ahead bias until
  you have proved otherwise. Real strategies are ugly.
```

A worked backtest skeleton, with the common mistakes annotated

PYTHON

```
"""
Minimal event-driven backtest skeleton (illustrative).
Deliberately simple so the LOGIC is visible rather than hidden
in a framework. The annotations marked MISTAKE show what most
retail backtests get wrong.

NOT a trading recommendation. Educational scaffolding only.
"""

import pandas as pd
import numpy as np


# ─────────────────────────────────────────────────────────
# 1. COST MODEL  -- build this FIRST, before any strategy
# ─────────────────────────────────────────────────────────
# MISTAKE #1: writing the strategy first and "adding costs
# later". Costs change which strategies are even viable, so
# they belong in the design loop from the beginning.

COST_CONFIG = {
    "brokerage_per_order": 20.0,       # flat, discount broker
    "gst_rate": 0.18,                  # on brokerage + fees
    "exchange_charge_rate": 0.0000325, # of turnover (verify)
    "sebi_fee_rate": 0.000001,         # of turnover (verify)
    "stamp_duty_rate": 0.00003,        # buy side only (verify)
    "stt_sell_rate": 0.001,            # verify by instrument
    "slippage_rate": 0.0005,           # 0.05% per side
}
# Every rate above is ILLUSTRATIVE and dated late 2026.
# Verify at nseindia.com / incometax.gov.in before use.


def round_trip_cost(entry_price, exit_price, qty, cfg):
    """Rupee cost of one complete round trip."""
    buy_turnover = entry_price * qty
    sell_turnover = exit_price * qty
    turnover = buy_turnover + sell_turnover

    brokerage = cfg["brokerage_per_order"] * 2
    exch = turnover * cfg["exchange_charge_rate"]
    sebi = turnover * cfg["sebi_fee_rate"]
    gst = (brokerage + exch + sebi) * cfg["gst_rate"]
    stamp = buy_turnover * cfg["stamp_duty_rate"]
    stt = sell_turnover * cfg["stt_sell_rate"]
    slip = turnover * cfg["slippage_rate"]

    return brokerage + exch + sebi + gst + stamp + stt + slip
```

```python
# ---------------------------------------------------------------
# 2. INDICATORS -- computed only from data available at the
#    time of the decision
# ---------------------------------------------------------------

def add_indicators(df):
    out = df.copy()
    out["sma20"] = out["close"].rolling(20).mean()
    out["sma50"] = out["close"].rolling(50).mean()
    out["sma200"] = out["close"].rolling(200).mean()

    prev_close = out["close"].shift(1)
    tr = pd.concat([
        out["high"] - out["low"],
        (out["high"] - prev_close).abs(),
        (out["low"] - prev_close).abs(),
    ], axis=1).max(axis=1)
    out["atr14"] = tr.rolling(14).mean()
    return out


# ---------------------------------------------------------------
# 3. SIGNAL -- evaluated on bar i, ACTED ON at bar i+1 open
# ---------------------------------------------------------------

def entry_signal(row):
    # MISTAKE #2: comparing row["close"] and then filling at
    # row["close"]. You cannot trade at a price you only knew
    # after it was set. Signals here are evaluated on bar i;
    # the engine below fills at bar i+1's OPEN.
    if np.isnan(row["sma200"]) or np.isnan(row["atr14"]):
        return False
    return (
        row["close"] > row["sma200"]
        and row["sma50"] > row["sma200"]
        and row["close"] < row["sma20"]
        and row["close"] > row["sma50"]
    )


# ---------------------------------------------------------------
# 4. ENGINE
# ---------------------------------------------------------------

def backtest(df, equity_start=500000.0, risk_frac=0.01,
             atr_stop_mult=1.5, atr_target_mult=2.5,
             max_bars=15, cfg=COST_CONFIG):
```

```
    df = add_indicators(df).reset_index(drop=True)
    equity = equity_start
    trades = []
    position = None

    for i in range(len(df) - 1):
        bar = df.iloc[i]
        nxt = df.iloc[i + 1]

        # ─── manage an open position ──────────────────
        if position is not None:
            position["bars_held"] += 1

            # MISTAKE #3: assuming the stop always fills at the
            # stop price. If the bar OPENS below the stop, you
            # fill at the OPEN, which is worse. Model that.
            stop_hit = nxt["low"] ≤ position["stop"]
            gap_through = nxt["open"] ≤ position["stop"]

            if stop_hit:
                fill = nxt["open"] if gap_through else position["stop"]
                reason = "gap_stop" if gap_through else "stop"
            elif nxt["high"] ≥ position["target"]:
                # MISTAKE #4: if BOTH the stop and the target are
                # inside the same bar, a daily-bar backtest cannot
                # know which came first. Resolving it in your
                # favour inflates results. Always resolve it
                # AGAINST yourself (stop first), as done here by
                # checking the stop before the target.
                fill = position["target"]
                reason = "target"
            elif position["bars_held"] ≥ max_bars:
                fill = nxt["open"]
                reason = "time_stop"
            else:
                fill = None
                reason = None

            if fill is not None:
                gross = (fill - position["entry"]) * position["qty"]
                cost = round_trip_cost(position["entry"], fill,
                                       position["qty"], cfg)
                net = gross - cost
                equity += net
                r_multiple = net / position["risk_rupees"]
                trades.append({
                    "entry_date": position["entry_date"],
                    "exit_date": nxt["date"],
```

```
                    "entry": position["entry"],
                    "exit": fill,
                    "qty": position["qty"],
                    "gross": gross,
                    "cost": cost,
                    "net": net,
                    "r": r_multiple,
                    "reason": reason,
                    "equity": equity,
                })
                position = None

        # ─── consider a new entry ──────────────────────
        if position is None and entry_signal(bar):
            atr = bar["atr14"]
            entry_px = nxt["open"] * (1 + cfg["slippage_rate"])
            stop_px = entry_px - atr_stop_mult * atr
            stop_dist = entry_px - stop_px
            if stop_dist ≤ 0:
                continue

            risk_rupees = equity * risk_frac
            qty = int(risk_rupees / stop_dist)
            # MISTAKE #5: allowing fractional or zero quantity.
            # If one unit exceeds the risk budget, take NO trade.
            if qty < 1:
                continue

            position = {
                "entry_date": nxt["date"],
                "entry": entry_px,
                "stop": stop_px,
                "target": entry_px + atr_target_mult * atr,
                "qty": qty,
                "risk_rupees": qty * stop_dist,
                "bars_held": 0,
            }

    return pd.DataFrame(trades), equity


# ──────────────────────────────────────────────────────────
# 5. METRICS
# ──────────────────────────────────────────────────────────

def metrics(trades, equity_start):
    if trades.empty:
        return {"trades": 0}
```

```python
    wins = trades[trades["net"] > 0]
    losses = trades[trades["net"] ≤ 0]

    win_rate = len(wins) / len(trades)
    avg_win_r = wins["r"].mean() if len(wins) else 0.0
    avg_loss_r = abs(losses["r"].mean()) if len(losses) else 0.0
    expectancy_r = (win_rate * avg_win_r) - ((1 - win_rate) * avg_loss_r)

    curve = pd.concat([
        pd.Series([equity_start]), trades["equity"]
    ], ignore_index=True)
    peak = curve.cummax()
    dd = (curve - peak) / peak
    max_dd = dd.min()

    gross_profit = wins["net"].sum() if len(wins) else 0.0
    gross_loss = abs(losses["net"].sum()) if len(losses) else 0.0

    return {
        "trades": len(trades),
        "win_rate": round(win_rate, 3),
        "avg_win_R": round(avg_win_r, 3),
        "avg_loss_R": round(avg_loss_r, 3),
        "expectancy_R": round(expectancy_r, 3),
        "profit_factor": round(gross_profit / gross_loss, 3)
                         if gross_loss else float("inf"),
        "max_drawdown_pct": round(max_dd * 100, 2),
        "total_costs": round(trades["cost"].sum(), 2),
        "cost_as_pct_of_gross": round(
            100 * trades["cost"].sum() / max(trades["gross"].sum(), 1e-9), 2),
        "final_equity": round(trades["equity"].iloc[-1], 2),
    }


# ──────────────────────────────────────────────────────────
# 6. THE SPLIT THAT MAKES THE RESULT MEAN ANYTHING
# ──────────────────────────────────────────────────────────
# MISTAKE #6 (the biggest one): tuning parameters on the whole
# dataset and reporting that as "the result". Split the data,
# develop on the in-sample portion ONLY, and touch the
# out-of-sample portion once.

def split_in_out_sample(df, in_sample_frac=0.6):
    cut = int(len(df) * in_sample_frac)
    return df.iloc[:cut].copy(), df.iloc[cut:].copy()
```

Six mistakes, one consequence

Each annotated mistake above inflates results independently. Combined, they routinely turn a losing strategy into a backtest showing a smooth 40% annual return. The order of magnitude is not an exaggeration — resolving same-bar stop/target ambiguity in your favour alone can swing a mean-reversion strategy from negative to strongly positive expectancy.

In-sample, out-of-sample and walk-forward

TEXT

```
THE THREE LEVELS OF VALIDATION
══════════════════════════════

LEVEL 1 – SIMPLE SPLIT
  ├──────── IN SAMPLE (60%) ────────┼── OUT OF SAMPLE (40%) ──┤
  Develop, tune and iterate here      Touch exactly ONCE

  Rule: you get ONE look at the out-of-sample data. If you
  look, adjust and look again, it is now in-sample and you
  have no held-out data left.

LEVEL 2 – WALK-FORWARD
  Train on window 1, test on the period immediately after.
  Roll forward. Repeat.

  ├──train 1──┼test 1┤
       ├──train 2──┼test 2┤
            ├──train 3──┼test 3┤
                 ├──train 4──┼test 4┤

  The concatenated test periods form an out-of-sample equity
  curve built from parameters that were only ever fitted on
  PRIOR data. This is much closer to how live trading works.

  Typical configuration: 3-year train, 1-year test, roll
  annually. Record whether the OPTIMAL parameters are stable
  across windows. If the best lookback is 14 in one window
  and 63 in the next, the parameter is fitting noise.

LEVEL 3 – PARAMETER SENSITIVITY SURFACE
  Do not report the best parameter set. Report the SURFACE.

  Lookback:      10     15     20     25     30     35
  Expectancy:  -0.02  +0.08  +0.31  +0.11  +0.04  -0.06

  A sharp spike at 20 surrounded by weak results is a sign
  of overfitting – you found the one setting that happened
  to work on this sample.

  Lookback:      10     15     20     25     30     35
  Expectancy:  +0.18  +0.22  +0.24  +0.23  +0.20  +0.16

  A broad plateau is what a real effect looks like. The
  precise parameter barely matters, which means you are
  capturing something structural rather than a coincidence.

ALWAYS PREFER A BROAD PLATEAU AT +0.20R OVER A SHARP PEAK
AT +0.50R. The plateau will survive contact with live data;
the peak will not.
```

Degrees of freedom and the overfitting budget

TEXT

```
COUNTING YOUR DEGREES OF FREEDOM
================================

Every tunable choice is a degree of freedom (DOF):
  - each indicator parameter (lookback, multiplier, threshold)
  - each rule included or excluded
  - the universe definition
  - the timeframe
  - the stop and target multiples
  - the regime filter thresholds

EXAMPLE COUNT for the specification earlier in this guide:
  SMA lookbacks (20, 50, 200)                    3 DOF
  ATR period (14)                                1 DOF
  Stop multiple (1.5)                            1 DOF
  Target multiple (2.5)                          1 DOF
  Time stop (15 bars)                            1 DOF
  VIX filter threshold (22)                      1 DOF
  Breakeven-move trigger (1.5 ATR)               1 DOF
  Rule set selection (6 entry conditions, each
    could have been included or not)             6 DOF
                                                 --------
  TOTAL                                         15 DOF

RULE OF THUMB
  You need a LOT of independent observations per degree of
  freedom. A commonly used heuristic is 30-50 trades per DOF
  as a bare minimum for the result to mean anything.

  15 DOF x 40 = 600 trades needed.

  If your backtest produced 85 trades, you have roughly 5.7
  trades per degree of freedom. You have not tested a
  strategy; you have drawn a curve through 85 points using
  15 parameters, which any undergraduate would recognise as
  guaranteed overfitting.

HOW TO REDUCE DOF
  - Fix parameters at conventional values and do not tune them
  - Remove rules rather than adding them; every rule you can
    delete without hurting results is a rule that was fitting
    noise
  - Test the SAME parameters across multiple instruments and
    multiple markets; if the edge is real it should appear,
    weakly, in several places
  - Prefer rules with an economic rationale over rules
    discovered by search
```

Why most backtests lie

Even a technically correct backtest can produce a result that will not repeat. The reasons are statistical rather than procedural, and they are the main source of the gap between simulated and live performance.

Multiple testing and p-hacking

TEXT

```
THE MULTIPLE TESTING PROBLEM, QUANTIFIED
═══════════════════════════════════════

Suppose you test 1,000 strategy variants on the same data.
Suppose NONE of them has any real edge.

If you use a 5% significance threshold, you expect
  1,000 x 0.05 = 50 variants
to appear "significantly profitable" purely by chance.

You will find them. You will be delighted. You will pick the
best one and trade it. It will fail.

HOW MANY TESTS DO YOU ACTUALLY RUN?
  People dramatically undercount. Every one of these is a test:
    - trying a different lookback                (x8)
    - trying a different stop multiple           (x6)
    - trying a different target multiple         (x6)
    - adding or removing a filter                (x4)
    - a different instrument                     (x5)
    - a different timeframe                      (x3)

  8 x 6 x 6 x 4 x 5 x 3 = 17,280 effective tests.

  Even if you only "looked at" 40 of these explicitly, your
  SEARCH SPACE was 17,280, and your brain pruned the rest.
  The multiple-testing correction applies to the search space,
  not to the number you wrote down.

THE DEFLATED EXPECTATION
  A practical correction: whatever your best backtested
  expectancy is, assume the live figure will be substantially
  lower. Some practitioners use a 50% haircut as a starting
  assumption. A strategy showing +0.40R in a tuned backtest
  should be planned around +0.20R or less live - and if it
  is not viable at +0.20R, do not trade it.

THE HONEST PROTOCOL
  1. Decide the hypothesis and parameters BEFORE testing.
  2. Write down how many variants you intend to test.
  3. Test them. Record ALL results, not just the best.
  4. If the best result is not dramatically better than the
     distribution of all results, you found noise.
  5. Validate the single chosen variant out of sample, once.
```

Regime dependence

Regime	Strategies that tend to work	Strategies that tend to fail
Sustained low-volatility uptrend	Trend following, buy-the-dip, covered calls, short volatility	Mean reversion on the short side, long volatility
Range-bound, choppy	Mean reversion, range trading, premium selling	Trend following and breakout — death by a thousand false signals
High-volatility decline	Long volatility, trend following on the short side, cash	Mean reversion ("buy the dip" into a bear market), all short-volatility structures
Transition / regime change	Almost nothing; this is where most strategies take their worst drawdowns	Everything that was working immediately before

⚠ **The regime trap**

A strategy developed and tested over 2020–2024 was developed in a particular Indian market regime. Its backtest tells you how it would have performed *in that regime*. If you cannot show the strategy working across multiple distinct regimes — including at least one sustained bear market and one extended sideways period — you do not know whether you have a strategy or a description of the recent past. Ten years of data covering several regimes is a reasonable minimum; it is also more data than most retail traders ever test on.

Capacity constraints

TEXT

CAPACITY – WHEN YOUR OWN SIZE DESTROYS THE EDGE

A backtest assumes you get the modelled price. In reality, your order participates in the market and moves it.

WORKED

Strategy trades a mid-cap stock
Average daily traded value (ADTV) Rs 12 crore
Your intended position Rs 5,00,000
Participation = 5,00,000 / 12,00,00,000 = 0.42% of ADTV
→ Impact cost modest. Fine.

Now you have grown the account and trade Rs 50,00,000
Participation = 4.2% of ADTV
→ You are now a meaningful part of the day's volume. Your buying pushes the price up as you enter and your selling pushes it down as you exit.
Realistic additional cost: 0.3–1.0% per side.

A strategy with a 0.8% gross edge per trade is destroyed.

THE SCALING CURVE

Position size (Rs)	Participation (% of ADTV)	Est. impact (per side)	Net edge (0.8% gross)
1,00,000	0.08%	0.02%	+0.76%
5,00,000	0.42%	0.06%	+0.68%
20,00,000	1.67%	0.22%	+0.36%
50,00,000	4.17%	0.65%	−0.50%
1,00,00,000	8.33%	1.40%	−2.00%

WHY THIS MATTERS EVEN IF YOU ARE SMALL

1. It tells you where a strategy's ceiling is. If the strategy stops working at Rs 20 lakh, it is a hobby, not a business.
2. It explains why genuine inefficiencies persist in small and illiquid corners – institutional money CANNOT exploit them, which is one of the few structural advantages a small retail account actually has.
3. It means you should test your strategy with a realistic impact model, not a flat slippage assumption.

Impact figures illustrative. NSE publishes impact cost data for index constituents – use the real numbers where available.

Pressure-testing a result before risking money

- [ ] **Remove the best 5% of trades.** If the strategy becomes unprofitable, it depends on a handful of outliers you cannot rely on repeating.
- [ ] **Double the cost assumption.** If it dies, your edge was within the error bars of your cost model.
- [ ] **Shift every entry and exit by one bar.** A real edge degrades gracefully; a fitted one collapses.
- [ ] **Perturb each parameter by ±20%.** Look for a plateau, not a peak.
- [ ] **Test on a different but related instrument.** A real effect should show up weakly elsewhere.
- [ ] **Test on a different period.** Split by regime, not just by date.
- [ ] **Randomise the trade order and recompute drawdown.** Your actual drawdown sequence was one draw from a distribution. Monte Carlo reshuffling shows you the range of drawdowns you could plausibly experience — it is usually much worse than the historical one.
- [ ] **Check the trade distribution.** Are profits spread across many trades, or concentrated in three days in March 2020?
- [ ] **Inspect your ten largest winners individually.** Data errors and look-ahead bias hide there.
- [ ] **Ask a sceptical friend to find the flaw.** Pay them if necessary. It is cheaper than the alternative.

PYTHON

```
"""
Monte Carlo reshuffle of a trade sequence.

Your historical maximum drawdown is ONE sample from the
distribution of drawdowns your strategy can produce. Reshuffling
the SAME trades in a different order shows the range.
Illustrative code; not a recommendation.
"""

import numpy as np


def monte_carlo_drawdowns(r_multiples, risk_frac=0.01,
                          n_sims=10000, seed=42):
    """r_multiples: array of per-trade results in R units."""
    rng = np.random.default_rng(seed)
    r = np.asarray(r_multiples, dtype=float)
    results = []

    for _ in range(n_sims):
        shuffled = rng.permutation(r)
        equity = 1.0
        curve = [equity]
        for x in shuffled:
            # fixed-fractional compounding: each trade risks
            # risk_frac of CURRENT equity
            equity *= (1.0 + risk_frac * x)
            curve.append(equity)
        curve = np.array(curve)
        peak = np.maximum.accumulate(curve)
        dd = (curve - peak) / peak
        results.append({
            "final": curve[-1],
            "max_dd": dd.min(),
        })

    finals = np.array([x["final"] for x in results])
    dds = np.array([x["max_dd"] for x in results])

    return {
        "median_final_multiple": float(np.median(finals)),
        "p05_final_multiple": float(np.percentile(finals, 5)),
        "p95_final_multiple": float(np.percentile(finals, 95)),
        "median_max_dd_pct": float(np.median(dds) * 100),
        "p95_worst_dd_pct": float(np.percentile(dds, 5) * 100),
        "prob_ruin_50pct": float((dds ≤ -0.50).mean()),
    }
```

```
# Typical interpretation:
#   If your historical max drawdown was -18% but the Monte
#   Carlo 5th percentile is -41%, then -41% is the number you
#   must be psychologically and financially prepared for.
#   The historical -18% was luck in the ORDERING of trades.
```

The Monte Carlo result is the number that matters

Most traders plan around their backtested drawdown. That figure is one draw from a distribution and is almost always optimistic. Plan around the 5th-percentile reshuffled drawdown instead. If you cannot survive that number — financially and emotionally — the strategy is too aggressively sized, regardless of what the equity curve looks like.

Position sizing and the risk of ruin

Position sizing is where most traders actually fail. A positive-expectancy strategy sized badly will destroy an account; a modest strategy sized well will survive long enough for the edge to compound. Sizing is not a detail of execution — it is the dominant determinant of whether you are still trading in two years.

The four sizing methods

Method	Rule	Strengths	Weaknesses
Fixed quantity	Always trade the same number of units/lots	Simple; no computation	Risk varies wildly with volatility and stop distance. A ₹200 stop and a ₹2,000 stop risk ten times as much on the same size. Effectively random risk control.
Fixed rupee	Always deploy the same rupee amount	Simple; caps notional exposure	Same problem — risk depends on stop distance, which you have not controlled.
Fixed fractional (percent risk)	Risk a fixed % of equity per trade; derive size from the stop distance	The standard. Risk is constant across trades. Position size scales down automatically in volatile conditions. Compounds naturally.	Requires a defined stop on every trade. Does not account for correlation between open positions.
Volatility-based (ATR)	Set the stop at a multiple of ATR, then apply fixed fractional	Adapts to each instrument's own volatility; makes positions comparable across instruments	ATR is backward-looking; it understates risk just before volatility expands.

Fixed fractional sizing, worked

TEXT

THE FORMULA

Position size (units) = (Equity x Risk%) / Stop distance

Always round DOWN. If the result is less than one tradeable unit, DO NOT TAKE THE TRADE.

WORKED ACROSS FOUR DIFFERENT TRADES

Account equity: Rs 5,00,000. Risk per trade: 1% = Rs 5,000.

Trade	Entry	Stop	Stop dist	Units	Notional	Risk
1	1,800	1,746	54.00	92	1,65,600	4,968
2	420	398	22.00	227	95,340	4,994
3	3,250	3,112	138.00	36	1,17,000	4,968
4	155	149.5	5.50	909	1,40,895	4,999

NOTICE

- Rupee risk is nearly identical (Rs 4,968 – 4,999) on all four, which is the entire point.
- NOTIONAL exposure varies from Rs 95,340 to Rs 1,65,600. That is fine. Notional is not risk; the stop distance is.
- The low-priced, tight-stop trade (#4) gets the largest unit count. That is correct, not dangerous – PROVIDED the stop is genuinely respected and the instrument does not gap.

NOW ADD VOLATILITY-BASED STOPS (ATR)

Instead of arbitrary stops, set stop = entry – 2 x ATR(14).

Trade	Entry	ATR14	Stop dist	Units	Rupee risk
1	1,800	31.00	62.00	80	4,960
2	420	9.40	18.80	265	4,982
3	3,250	74.00	148.00	33	4,884
4	155	4.10	8.20	609	4,994

Now your stop distance is proportional to how much each instrument actually moves. A volatile instrument gets a wider stop and a correspondingly smaller position. Two trades in two very different instruments now represent the same amount of risk AND the same probability of being stopped out by noise.

THIS IS THE CORRECT DEFAULT FOR MOST RETAIL STRATEGIES.

Portfolio heat

TEXT

```
PORTFOLIO HEAT - THE RISK YOU FORGET TO COUNT
═══════════════════════════════════════════════
HEAT = the sum of the risk on all currently open positions,
       as a % of equity.

  5 open positions at 1% risk each = 5% heat.

WHY THIS MATTERS
  If all five positions are long equity in a correlated
  market, they are not five independent bets. They are one
  bet with five tickers. A market-wide 4% down day can hit
  all five stops simultaneously.

  Your "1% risk per trade" becomes a 5% day.
  Three such days in a month = -15%.

CORRELATION-ADJUSTED HEAT
  Crude but useful adjustment:

  Effective heat = Nominal heat x sqrt(1 + (n-1) x rho)/sqrt(n)
                   ... for n equally-sized positions with
                   average pairwise correlation rho

  n=5, rho=0.0  (independent):
    5% x sqrt(1)/sqrt(5)  ...  effective ~2.2%  (diversified)
  n=5, rho=0.7  (typical for Indian large caps in a selloff):
    effective heat approaches the full 5%

  In a crisis, correlations go to 1. Plan for rho = 1.

PRACTICAL HEAT LIMITS
  Conservative          : 3% maximum heat
  Moderate              : 5% maximum heat
  Aggressive            : 8% maximum heat
  Reckless              : no limit at all (most retail)

  With a 5% heat cap and 1% per trade, you may hold at most
  5 positions. Wanting a sixth means closing one first.
  This constraint alone prevents a large category of blow-ups.

SECTOR AND FACTOR HEAT
  Also cap exposure by sector. Five positions in five
  different banks is one position in "banks".
```

Risk of ruin

TEXT

RISK OF RUIN — THE TABLE EVERY TRADER SHOULD SEE ONCE
═══

"Ruin" here means a drawdown large enough that recovery is
impractical. We will use −50% as the ruin threshold, because
recovering from −50% requires +100%.

SIMULATION SETUP
  Strategy expectancy +0.20R, win rate 45%,
  average win 2.0R, average loss 1.0R.
  Fixed fractional sizing. 500 trades.
  10,000 Monte Carlo paths.

Risk per trade	Prob. of −50% DD	Median max DD	Median final	5th pctile final
0.5%	0.2%	−8.4%	1.65x	1.18x
1.0%	1.9%	−15.8%	2.69x	1.33x
2.0%	14.1%	−29.1%	7.02x	1.42x
3.0%	33.8%	−40.3%	17.1x	1.11x
5.0%	68.2%	−59.7%	78.5x	0.31x
10.0%	95.6%	−86.1%	410x	0.004x

READ THE 5% ROW VERY CAREFULLY
  Median outcome: 78.5x your starting capital. Spectacular.
  5th percentile outcome: you have 31% of what you started
  with — a 69% loss.
  Probability of a 50%+ drawdown along the way: 68%.

  A STRATEGY WITH A GENUINE POSITIVE EDGE RUINS TWO THIRDS OF
  THE PEOPLE WHO TRADE IT AT 5% RISK PER TRADE.

  The median looks wonderful because the distribution is
  wildly skewed. The median is not your expected experience;
  the whole distribution is. And most people quit at −40%,
  so they never reach the median.

THE 2% vs 5% COMPARISON, SPELLED OUT
  At 2% risk: 14% chance of a 50% drawdown
  At 5% risk: 68% chance of a 50% drawdown
  The risk per trade went up 2.5x.
  The probability of ruin went up nearly 5x.

  Risk of ruin is HIGHLY NON-LINEAR in position size. This
  is the single most important fact in this entire guide.

AND WITH A NEGATIVE EDGE
  Expectancy −0.05R (a slightly losing strategy, which is
  what most retail strategies are after costs):

```
  Risk per     Prob of
  trade        −50% DD over 500 trades
  ─────────    ──────────────────────────

    1.0%              78%
    2.0%              94%
    5.0%             100%

  With a negative edge, ruin is not a risk. It is a schedule.
  Position sizing only determines how long it takes.

Figures from illustrative simulation. The SHAPE is robust.
```



The same strategy, the same edge, six different sizing choices. Going from 2% to 5% risk per trade raises the probability of a 50% drawdown from 14% to 68%.

Kelly and why you should use a fraction of it

TEXT

THE KELLY CRITERION

Kelly gives the position size that maximises the LONG-RUN GEOMETRIC GROWTH RATE of capital.

For a bet with win probability W, payoff ratio P (win/loss):

$$f^* = W - (1 - W) / P$$

WORKED

W = 0.45, P = 2.0

$f^* = 0.45 - 0.55/2.0 = 0.45 - 0.275 = 0.175$

Kelly says risk 17.5% of capital per trade.

WHY YOU MUST NOT DO THAT

1. KELLY ASSUMES YOU KNOW W AND P EXACTLY. You do not. You have estimates from a finite sample, contaminated by overfitting. If your true W is 0.40 rather than 0.45, the true Kelly is 0.10 - and betting 0.175 is OVER-BETTING, which produces NEGATIVE long-run growth. Kelly is brutally asymmetric: under-betting costs you a little growth, over-betting destroys you.
2. KELLY TOLERATES ENORMOUS DRAWDOWNS. At full Kelly the expected maximum drawdown approaches 50-60%+. No human trades through that calmly.
3. KELLY ASSUMES THE EDGE IS STABLE. Market edges decay.
4. KELLY ASSUMES INDEPENDENT BETS. Your positions are correlated.

FRACTIONAL KELLY

Common practice is to use a small fraction of full Kelly:

Fraction	Position size	Growth vs full Kelly	Max DD
Full	17.5%	100%	~55%
1/2 Kelly	8.8%	75%	~33%
1/4 Kelly	4.4%	44%	~19%
1/8 Kelly	2.2%	23%	~10%
1/16 Kelly	1.1%	12%	~5%

Note the shape: half Kelly gives you 75% of the growth for 60% of the drawdown. Quarter Kelly gives you 44% of the growth for a third of the drawdown.

IN PRACTICE, 1/8 TO 1/4 KELLY IS WHERE SERIOUS PRACTITIONERS OPERATE, which lands at roughly 1-4% risk per trade - the same place that simple fixed-fractional reasoning arrives at.

```
Kelly is therefore most useful not as a sizing rule but as
a CEILING: if your sizing exceeds half Kelly, you are
over-betting even under the most favourable assumptions
about your own edge.
```

PYTHON

```
"""
Position sizing helpers. Illustrative; not a recommendation.
"""

import math


def fixed_fractional_size(equity, risk_frac, entry, stop,
                          lot_size=1):
    """Units to trade, rounded DOWN to whole lots.
    Returns 0 if one lot would exceed the risk budget."""
    stop_distance = abs(entry - stop)
    if stop_distance ≤ 0:
        raise ValueError("stop must differ from entry")

    risk_rupees = equity * risk_frac
    raw_units = risk_rupees / stop_distance
    lots = math.floor(raw_units / lot_size)
    units = lots * lot_size

    # If a single lot already risks more than the budget,
    # the correct answer is NO TRADE - not "round up".
    if units == 0:
        return 0, 0.0
    return units, units * stop_distance


def atr_stop_size(equity, risk_frac, entry, atr, atr_mult=2.0,
                  lot_size=1):
    stop = entry - atr_mult * atr
    return fixed_fractional_size(equity, risk_frac, entry,
                                 stop, lot_size)


def kelly_fraction(win_rate, payoff_ratio):
    """Full Kelly. Use a FRACTION of this, never the whole."""
    if payoff_ratio ≤ 0:
        return 0.0
    f = win_rate - (1.0 - win_rate) / payoff_ratio
    return max(f, 0.0)


def sizing_sanity_check(equity, risk_frac, open_positions_risk,
                        max_heat=0.05):
    """Reject a new trade that would breach portfolio heat."""
    proposed_heat = (open_positions_risk + equity * risk_frac) / equity
    return {
        "proposed_heat_pct": round(proposed_heat * 100, 2),
```

```
        "max_heat_pct": round(max_heat * 100, 2),
        "allowed": proposed_heat ≤ max_heat,
    }


# Example
if __name__ == "__main__":
    units, risk = atr_stop_size(
        equity=500000, risk_frac=0.01,
        entry=1800.0, atr=31.0, atr_mult=2.0)
    print(units, round(risk, 2))
    # Kelly ceiling check
    k = kelly_fraction(win_rate=0.45, payoff_ratio=2.0)
    print("full kelly:", round(k, 4),
          "| quarter kelly:", round(k / 4, 4))
```

The sizing rules that are non-negotiable

Risk no more than 1–2% of trading capital per trade. Cap total portfolio heat at 5%. Never increase size to recover a loss. Never size by "how confident I feel". Never let one position exceed 20% of the account's notional regardless of stop distance. If you break these, nothing else in this guide can save you.

Stop losses done properly

A stop loss is a pre-commitment device. Its purpose is to make the decision to exit while you are rational, so that your irrational self cannot override it. Most of what goes wrong with stops is not about placement — it is about honouring them.

Logical versus arbitrary stops

Stop type	Definition	Assessment
Percentage stop	Fixed % below entry, e.g. 5%	Arbitrary. 5% is noise in a volatile small cap and a major move in a stable large cap. Use only as a crude cap.
Rupee stop	"I will not lose more than ₹10,000"	Arbitrary and backwards — it places the stop based on your wallet rather than on the market's structure. It does belong in sizing, not in placement.
Volatility (ATR) stop	Entry − k × ATR	**The sensible default.** Adapts to each instrument. A 2×ATR stop has roughly comparable "noise-out" probability across instruments.
Structural stop	Below a swing low, a consolidation base, a prior breakout level	Good when the level is objectively definable. Risk: everyone sees the same level, so it attracts stop-hunting order flow.
Thesis stop	Exit when the reason for the trade is no longer true	Intellectually the purest. Requires the thesis to have been written down precisely enough to be falsified, which is rare.
Time stop	Exit after N bars regardless of P&L	Underused and valuable. If the move has not happened in the expected window, the hypothesis is not playing out. Frees capital and caps exposure.
Trailing stop	Stop follows price up by a fixed amount or ATR multiple	Good for trend strategies; converts open profit into realised profit. Trails too tightly and you exit on noise; too loosely and you give back most of the gain.
Mental stop	"I will exit if it hits X" — not entered in the system	**Usually not a stop at all.** See below.

Why a mental stop is usually not a stop

A stop exists to remove the decision from your stressed self. A mental stop hands the decision right back at the exact moment you are least capable of making it — when the position is losing, your pulse is up, and every narrative in your head is arguing for one more minute. Experienced discretionary traders with a long track record can run mental stops. Beginners who use mental stops are, with high reliability, running no stop at all. Place the order.

Where not to place a stop

- **At a round number.** 1,800.00, 25,000, 500 — clusters of stops sit there and price is drawn through them.
- **Exactly at the obvious swing low.** Everyone's stop is there. Place it a volatility-scaled distance beyond, or accept being stopped on noise.
- **Inside the instrument's normal daily range.** If ATR is 30 points and your stop is 12 points away, you are not managing risk; you are paying for a coin flip with costs attached.
- **At a distance determined by your position size.** This is backwards. Decide the stop from the market, then derive the size from the stop.
- **Nowhere, because "it is a quality company".** Quality is a reason to invest, not a reason to carry an unbounded loss on a leveraged trade.

TEXT

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STOP PLACEMENT WORKED EXAMPLE
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Instrument: a hypothetical large-cap share
Entry  Rs 1,800
ATR(14) Rs 31.00
Recent swing low Rs 1,742
Account Rs 5,00,000, risk 1% = Rs 5,000

CANDIDATE STOPS
A. 3% stop          → 1,746.00   distance 54.00
B. 2 x ATR stop     → 1,738.00   distance 62.00
C. Swing low        → 1,742.00   distance 58.00
D. Swing low
   minus 0.5 ATR    → 1,726.50   distance 73.50
E. Rs 10,000 cap    → depends on size; circular

RESULTING POSITION SIZES (risk Rs 5,000)
A. 5,000 / 54.00   =  92 units, notional Rs 1,65,600
B. 5,000 / 62.00   =  80 units, notional Rs 1,44,000
C. 5,000 / 58.00   =  86 units, notional Rs 1,54,800
D. 5,000 / 73.50   =  68 units, notional Rs 1,22,400

WHICH IS BEST?
  D is the most defensible: it respects the market structure
  (the swing low) AND adds a volatility buffer so that normal
  noise does not take you out at the obvious level where
  every other stop sits.

  Its cost is a smaller position. That is the correct
  trade-off and the one most people get wrong - they tighten
  the stop to justify a bigger position, which maximises the
  chance of being stopped out on noise while also maximising
  the damage when the stop fails in a gap.

  WIDER STOP + SMALLER SIZE beats TIGHTER STOP + BIGGER SIZE
  for almost every retail strategy.

NOW THE REALITY CHECK
  What if the share gaps to 1,600 on news?
  Stop at D (1,726.50) does not protect you.
  Fill around 1,598. Loss = (1,800-1,598) x 68 = Rs 13,736
  That is 2.75R, not 1R.
  Plan for it: see the gap maths in the market structure
  section. Your sizing must survive a 3R adverse outcome.
```

Trailing stops: the give-back problem

TEXT

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TRAILING STOP COMPARISON ON THE SAME TRADE SEQUENCE
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Entry at 1,800. The trade runs to 2,100 and then reverses
to 1,850 before you would otherwise have exited.
ATR at entry: 31.00

METHOD                          EXIT    PROFIT/UNIT   % CAPTURED
──────────────────────────────  ──────  ────────────  ──────────
No trail, fixed target 2.5R     1,955       155          52%
                                (hit on the way up)
Trail at 1 x ATR (31)           2,062       262          87%
Trail at 2 x ATR (62)           2,024       224          75%
Trail at 3 x ATR (93)           1,990       190          63%
Trail at 10% below high         1,890        90          30%
Chandelier (3 ATR from high)    1,990       190          63%
No trail, exit on reversal
  signal                        1,862        62          21%

THE TRADE-OFF
  A TIGHT trail captures more of THIS move but would have
  exited many earlier trades prematurely on normal pullbacks.
  A LOOSE trail gives back more at the top but stays in
  genuine trends.

  You cannot evaluate a trail on one trade. You must test it
  across the full trade distribution, where the question
  becomes: does the extra profit captured on the big winners
  exceed the profit lost by being shaken out of medium
  winners?

  For trend strategies the answer is usually YES for a 2.5-3.5
  ATR trail and NO for anything tighter than 1.5 ATR.
  For mean-reversion strategies, trailing stops are usually
  harmful - the strategy's premise is a bounded move, so a
  fixed target is more appropriate.

TEST YOUR TRAIL. Do not adopt one because it sounds prudent.
```

The move to breakeven

Moving the stop to the entry price once a trade is up 1–1.5R feels like free risk reduction. It is not free: it increases the probability of a scratch exit on a normal pullback, which truncates winners. Test it. For many trend strategies it reduces expectancy. For strategies with a high proportion of trades that reverse through entry, it helps. The answer is empirical, not obvious.

Drawdowns: the thing that actually ends trading careers

Almost nobody quits trading because of a single bad trade. They quit during a drawdown — a long, grinding period where nothing works, the equity curve bleeds, and every day's decision is made under the weight of accumulated loss. Understanding drawdown maths and preparing for it in advance is the difference between a bad quarter and the end.

Recovery arithmetic

TEXT

```
THE ASYMMETRY OF LOSS
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  Required gain to recover = 1/(1 − loss) − 1

Drawdown   Gain needed    Interpretation
to recover
─────────  ────────────   ──────────────────────────────────

   −5%         +5.3%      Trivial
  −10%        +11.1%      Routine
  −20%        +25.0%      A bad year
  −25%        +33.3%      Serious
  −30%        +42.9%      You are questioning everything
  −40%        +66.7%      Most people quit here
  −50%       +100.0%      You must DOUBLE to get back
  −60%       +150.0%
  −70%       +233.3%
  −80%       +400.0%      Practically unrecoverable
  −90%       +900.0%      Over

THE COMPOUNDING INSIGHT
  Capital destroyed is also TIME destroyed. If your strategy
  genuinely produces 15% a year:
    recovering −20% takes about 1.6 years
    recovering −40% takes about 3.7 years
    recovering −50% takes about 5.0 years
    recovering −70% takes about 8.6 years

  And during the recovery you earn nothing — you are merely
  returning to where you already were. A −50% drawdown on a
  15%-a-year strategy costs you five years of compounding.

  This is why risk control is not conservatism. It is the
  mathematically optimal behaviour for anyone who intends
  to still be trading in five years.
```



A plausible equity curve for a strategy with a genuine edge. Note that the duration of the flat period — a year with no progress — is usually harder to endure than the depth of the decline.

Expected drawdown for a given strategy

TEXT

ESTIMATING THE DRAWDOWN YOU SHOULD EXPECT

A useful approximation for the expected maximum drawdown, in R units, over N trades:

  Expected max losing streak (consecutive losses)
      approximately = ln(N) / -ln(L)

  where L = loss rate = 1 - win rate

WORKED, three strategies, N = 500 trades

Strategy	Win rate	L	ln(500)	-ln(L)	Expected max streak
Trend	35%	0.65	6.215	0.4308	14.4
Balanced	50%	0.50	6.215	0.6931	9.0
Mean rev.	70%	0.30	6.215	1.2040	5.2

TRANSLATE INTO MONEY (1% risk per trade)

  Trend following: a 14-trade losing streak at -1R each
  = roughly -13% to -14% from that streak ALONE, before
  counting the partial losses and scratch trades around it.

  And a 14-loss streak at 100 trades a year means you will
  spend about seven weeks losing on every single trade.

  CAN YOU DO THAT WITHOUT CHANGING YOUR BEHAVIOUR?
  Most people cannot. That is why trend following, despite
  having the best expectancy profile on paper, is abandoned
  by most retail traders who attempt it.

A MORE COMPLETE DRAWDOWN ESTIMATE

  Max drawdown is driven by streaks AND by the size of
  losses AND by the ordering. Use the Monte Carlo reshuffle
  from the backtesting section rather than this formula for
  anything you will act on. The formula is for intuition.

THE PLANNING NUMBER

  Take your backtested max drawdown.
  Take the Monte Carlo 5th percentile.
  Take the larger. Multiply by 1.5.
  THAT is the number to plan your capital and your
  psychology around.

Is the edge gone, or is this normal variance?

Signal	Suggests normal variance	Suggests the edge is gone
Drawdown depth	Within 1.5× the backtested/Monte Carlo maximum	Beyond 1.5× the worst expected drawdown
Drawdown duration	Within the historical range of flat periods	Materially longer than any historical flat period
Trade statistics	Win rate and payoff ratio within their historical confidence bands	Win rate has shifted outside its band and stayed there for 50+ trades
Execution quality	Slippage and fills in line with the model	Slippage consistently worse than modelled — liquidity has changed
Rule adherence	You have followed every rule	You have been deviating — in which case the strategy is not what is failing
Market structure	Nothing fundamental has changed	A regulatory change, a product change, a structural shift in the instrument
Crowding	No evidence others are doing this	The approach has become widely publicised and popular

⚠ **The hardest judgement in trading**

Distinguishing "my edge has decayed" from "I am in a normal bad patch" cannot be done reliably in real time with retail sample sizes. This is why the decision must be *pre-committed*: you write the kill criteria into the specification before you trade, so that the decision is made by the plan rather than by your state of mind during the drawdown.

A drawdown response protocol

1. **At −5%: nothing changes**

   This is normal. Do not adjust anything. Do not "tighten up". Continue executing the plan exactly.

2. **At −10%: audit execution, not strategy**

   Review the last 20 trades in your journal. Did you follow every rule? Count the deviations. If there are deviations, the problem is you, not the strategy — fix adherence and continue.

3. **At −15%: halve position size**

Risk per trade goes from 1% to 0.5%. This is not a loss of nerve; it reduces the chance of a bad patch becoming unrecoverable, and it buys you more trades per unit of capital to re-establish whether the edge is intact.

4 **At −20%: stop and do a full review**
Two weeks away from the market. Re-run the strategy's statistics on the live trades. Compare against the backtest confidence bands. Check the kill criteria. Write a conclusion before returning.

5 **At −25%: stop trading live**
Back to paper trading until you can demonstrate 30 trades at the specified expectancy. This is the pre-committed line you wrote down when you were calm.

6 **At any point: if you have deviated from the plan twice in a week, stop immediately**
Deviation under pressure is the leading indicator of the large loss. Pause regardless of the P&L.

✕ **What you must never do in a drawdown**
Increase position size to recover faster. Widen stops to avoid taking losses. Abandon the strategy for a new one you found last week. Add a correlated position to "average down" the overall book. Move money from savings into the trading account. Trade a new instrument you have not tested. Each of these converts a recoverable drawdown into a terminal one, and each feels entirely reasonable at the time.

Costs and taxes as a performance drag

Costs are deterministic. Your edge is probabilistic. That asymmetry means costs compound against you with certainty while your edge only works on average, which is why cost control is the highest-confidence improvement available to any trader.

⚠ **Every rate below is illustrative and dated**
Brokerage models, STT rates, exchange charges, stamp duty and tax provisions have all changed in recent years, and the Income-tax Act 2025 restructured and renumbered provisions with effect from 1 April 2026. Figures here are plausible for **late 2026 / FY 2026–27** and exist to teach the *structure* of the drag. Verify current rates at **nseindia.com**, **incometax.gov.in** and in your broker's published tariff.

The cost stack for Indian equity trading

Charge	Delivery equity	Intraday equity	Futures	Options
Brokerage	Often zero or flat	Flat per order, or a % capped at a flat amount	Flat per order	Flat per order
STT	Both sides, on turnover	Sell side, on turnover	Sell side, on turnover	Sell side on **premium**; on **settlement value** if exercised
Exchange transaction charge	On turnover	On turnover	On turnover	On **premium** turnover
SEBI turnover fee	Per crore of turnover	Same	Same	Same
Stamp duty	Buy side	Buy side	Buy side	Buy side
GST	18% on brokerage + exchange + SEBI fees	Same	Same	Same
DP charges	Per scrip per day on sell (delivery)	—	—	—
Spread & impact	Yes	Yes	Yes	Yes — largest item in illiquid strikes

The worked example: a genuine 0.4% edge destroyed by frequency

TEXT

```
A STRATEGY WITH A REAL EDGE, RUN AT FOUR FREQUENCIES
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Assume a genuine gross edge of 0.40% of position notional
per round trip. This is a GOOD strategy - most are not
this good.

Position notional per trade: Rs 2,00,000
Gross profit per trade: 2,00,000 x 0.004 = Rs 800

COST PER ROUND TRIP (intraday equity, illustrative rates)
  Turnover (buy + sell)              Rs 4,00,000
  Brokerage: 0.03% capped at Rs 20
             per order x 2                Rs  40.00
  STT: 0.025% of sell turnover
       2,00,000 x 0.00025                 Rs  50.00
  Exchange charge: 0.00325% of turnover
       4,00,000 x 0.0000325               Rs  13.00
  SEBI fee: Rs 10 per crore
       4,00,000 x 0.000001                Rs   0.40
  Stamp duty: 0.003% buy side
       2,00,000 x 0.00003                 Rs   6.00
  GST 18% on (40 + 13 + 0.40)             Rs   9.61
  SLIPPAGE 0.03% per side
       4,00,000 x 0.0003                  Rs 120.00
                                          ──────────
  TOTAL COST PER ROUND TRIP               Rs 239.01

NET PER TRADE = 800.00 - 239.01 = Rs 560.99
COST AS % OF GROSS PROFIT = 29.9%

Nearly a third of a genuinely good edge goes to friction.

NOW VARY THE EDGE AND THE FREQUENCY
──────────────────────────────────

Gross edge   Gross/trade   Cost/trade   Net/trade   Net edge
──────────   ──────────   ──────────   ─────────   ────────
  0.40%         800.00       239.01       560.99     0.280%
  0.25%         500.00       239.01       260.99     0.130%
  0.15%         300.00       239.01        60.99     0.030%
  0.12%         240.00       239.01         0.99     0.000%  ←BE
  0.10%         200.00       239.01       -39.01    -0.020%
  0.05%         100.00       239.01      -139.01    -0.070%

THE BREAK-EVEN EDGE IS 0.12% PER ROUND TRIP.
Any strategy with a gross edge below that loses money with
certainty, no matter how clever it is.

NOW ANNUALISE IT ON A Rs 5,00,000 ACCOUNT
```

Trades/ year	Gross P&L (@0.40%)	Costs	Net P&L	Return on Rs 5,00,000
25	20,000	5,975	14,025	+2.8%
50	40,000	11,951	28,049	+5.6%
100	80,000	23,901	56,099	+11.2%
250	2,00,000	59,753	1,40,248	+28.0%
500	4,00,000	1,19,505	2,80,495	+56.1%
1000	8,00,000	2,39,010	5,60,990	+112.2%

"SO HIGH FREQUENCY IS BETTER?" - NO, AND HERE IS WHY

The table above assumes the 0.40% edge is AVAILABLE 1,000 times a year at Rs 2 lakh notional. It is not. Edges have CAPACITY. The more often you trade, the lower-quality the average opportunity, because you are forced to take the marginal signal as well as the good one.

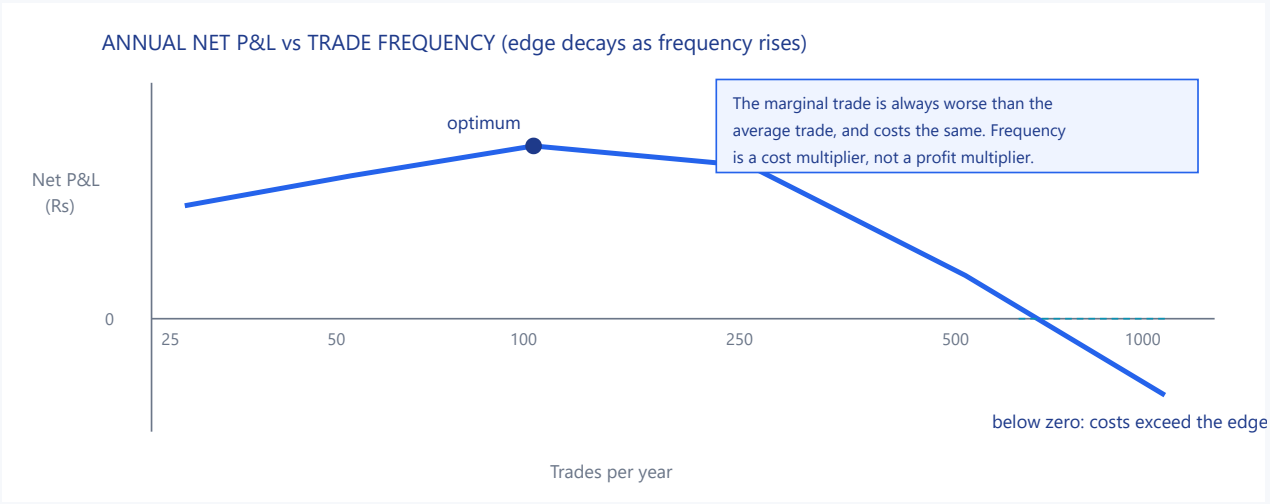
REALISTIC: edge DECAYS with frequency.

Trades/ year	Realistic gross edge	Gross/ trade	Cost/ trade	Net/ trade	Annual net
25	0.80%	1,600.00	239.01	1,360.99	34,025
50	0.55%	1,100.00	239.01	860.99	43,050
100	0.40%	800.00	239.01	560.99	56,099
250	0.22%	440.00	239.01	200.99	50,248
500	0.14%	280.00	239.01	40.99	20,495
1000	0.09%	180.00	239.01	-59.01	-59,010

THE OPTIMUM IS IN THE MIDDLE, AND IT IS MUCH LOWER FREQUENCY THAN MOST RETAIL TRADERS OPERATE AT.

Beyond roughly 100-250 trades a year on a Rs 5 lakh account, additional trading destroys value. The marginal trade is worse than the average trade AND costs the same.

All rates illustrative. As at late 2026 - verify at nseindia.com and incometax.gov.in.



Net profit against trade frequency on a ₹5 lakh account, assuming the available edge per trade decays as you take more marginal signals. The optimum is far lower than most retail traders operate at.

✦ **The highest-return activity available to a retail trader**

Not finding a better signal. Reducing trade frequency and improving fill quality. Both are under your direct control, both are deterministic, and both work immediately. Most people spend their effort on the signal, which is the part they have least ability to improve.

Taxation of trading income in India

⚠ **This is a summary of structure, not tax advice**

Indian trading taxation is genuinely complex and has changed. The Income-tax Act 2025 took effect from 1 April 2026 with renumbered provisions and the "tax year" concept. Capital gains rates and holding periods for equity were revised in 2024. **As at FY 2026–27, verify everything at incometax.gov.in** and engage a chartered accountant. Do not plan around this table.

Activity	Head of income (typical treatment)	Rate	Loss set-off and carry-forward
Equity delivery, held long-term	Long-term capital gains	Concessional LTCG rate, with an annual exemption threshold	LTCL against LTCG only; carry forward for a limited number of years
Equity delivery, held short-term	Short-term capital gains	Concessional STCG rate for STT-paid equity	STCL against STCG or LTCG; carry forward subject to conditions
Intraday equity	**Speculative business income**	Slab rate	Speculative losses can be set off only against speculative income; carry forward for a limited period
F&O (exchange-traded derivatives)	**Non-speculative business income**	Slab rate	Can be set off against most other heads (not salary); carry forward subject to filing the return on time
Delivery trading done as a business	Business income, if the facts support it	Slab rate	As non-speculative business

TEXT

```
WHAT BUSINESS-INCOME TREATMENT ACTUALLY MEANS
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1. YOU CAN DEDUCT EXPENSES
   Brokerage, STT (as a business expense where allowed),
   exchange charges, GST on services, internet, a proportion
   of electricity, depreciation on a computer, data
   subscriptions, advisory fees, demat charges, and interest
   on money genuinely borrowed for the business.
   Keep invoices. Keep a separate bank account.

2. YOUR GAINS ARE TAXED AT SLAB RATES
   For a software engineer already in a high slab, trading
   profits may be taxed at the top marginal rate plus
   surcharge and cess. Compare that to the concessional
   LTCG treatment on long-held equity. The tax wedge alone
   is a meaningful argument for investing over trading.

3. TURNOVER AND AUDIT
   "Turnover" for F&O has a SPECIFIC definition that is not
   the notional value of your trades. Broadly, the sum of
   absolute profits and losses (and, in some interpretations,
   premium received on options written) rather than contract
   value. Above specified turnover thresholds - and in
   certain loss-reporting situations - a TAX AUDIT may be
   required, which has its own cost and deadline.
   This catches people who traded a small account actively
   and discover they have an audit obligation.

4. ADVANCE TAX
   Business income attracts advance tax in instalments
   through the year. Interest is charged for shortfalls.

5. LOSSES ARE ONLY CARRIED FORWARD IF YOU FILE ON TIME
   A loss-making year still requires a timely return in
   order to preserve the carry-forward. Many traders skip
   filing in a loss year and permanently forfeit the set-off.

6. YOU WILL NEED THE RIGHT ITR FORM
   Business income generally means a different ITR form from
   the simple salaried one, with a profit-and-loss statement
   and balance sheet schedule.

A WORKED SKETCH (ILLUSTRATIVE ONLY)
──────────────────────────────────

Salary income                    Rs 24,00,000
F&O gross profit                 Rs  3,00,000
Less deductible expenses
  Brokerage and charges          Rs    85,000
```

```
  Data and software                Rs    18,000
  Internet (business share)        Rs     9,000
  Laptop depreciation              Rs    15,000
  Professional fees (CA)           Rs    25,000
                                   ____________
Net F&O business income            Rs  1,48,000

Total taxable income               Rs 25,48,000
  taxed per the applicable regime and slab;
  the F&O component is taxed at the MARGINAL rate.

If instead the F&O result had been a LOSS of Rs 1,52,000,
that non-speculative loss could generally be set off against
other income (not salary) in the same year, with the balance
carried forward - PROVIDED the return is filed within the
due date.

Entirely illustrative. As at FY 2026-27 - verify at
incometax.gov.in and consult a chartered accountant.
```

> **The after-tax comparison nobody runs**
>
> A trading strategy returning 18% gross, taxed at a 31.2% marginal rate, nets about 12.4%. A low-cost index fund held long term, returning 11% gross with concessional capital-gains treatment and no annual tax event on unrealised gains, can net a comparable or better figure with a fraction of the effort and risk. Run this comparison honestly before deciding that trading is worth your time.

The trading journal

The journal is the single highest-value habit available to a trader, and the one most consistently skipped. Without it you have no data about your own behaviour, which means you cannot distinguish a strategy problem from an execution problem, and you cannot improve anything that is not measured.

What to record, and when

PRE-TRADE (BEFORE THE ORDER)

Hypothesis, setup tag, entry trigger, planned stop, planned target, position size calculation, R value in rupees, portfolio heat after this trade, conviction level, your emotional state, and anything about the market context. Written *before* you can be influenced by the outcome.

POST-TRADE (AFTER THE EXIT)

Actual fills, actual costs, R-multiple, exit reason, whether every rule was followed, what you would do differently, and — critically — a separate assessment of **decision quality** versus **outcome**. A good decision can lose money. A bad decision can make money.

Separate decision quality from outcome

Grade every trade twice: once on whether you followed your process (A–F), once on the financial result. Over time, the correlation between these two grades is the most informative statistic you will ever compute about yourself. If your F-graded decisions are profitable, you are being rewarded for bad behaviour, which is the most dangerous situation in trading.

A full journal template

TEXT

```
TRADE JOURNAL - PRE-TRADE ENTRY
================================
TRADE ID      : 2026-0147
DATE/TIME     : 2026-10-21  09:42
INSTRUMENT    : [index futures / ETF / stock - specify]
DIRECTION     : LONG
STRATEGY      : Index Pullback Continuation v0.3
SETUP TAG     : PULLBACK-TO-20SMA-IN-UPTREND

--- THE HYPOTHESIS (write this BEFORE entering) ---
Why this trade, in one sentence:
  "Medium-term uptrend intact (close above 200SMA, 50SMA above
   200SMA); price pulled back below the 20SMA for 3 sessions
   and closed up yesterday; VIX at 13.8, below the 22 filter."

What would make me wrong:
  "A close below the 50SMA, or a close below the 200SMA
   (regime exit)."

--- RULE CHECKLIST (all must be YES) ---
  E1 close > 200SMA ............ YES
  E2 50SMA > 200SMA ............ YES
  E3 close < 20SMA ............. YES
  E4 close > 50SMA ............. YES
  E5 close > prior close ....... YES
  E6 VIX < 22 .................. YES  (13.8)
  F1 no event in next 3 days  ... YES
  F2 open positions < 2 ........ YES  (1 open)
  F3 heat after this trade ..... 2.0%  (limit 4%) OK

--- NUMBERS ---
  Account equity        Rs 5,00,000
  Risk per trade        1.0%  = Rs 5,000  (1R)
  Planned entry         25,040 (next open)
  ATR(14)               186 points
  Stop (1.5 x ATR)      24,761  -> distance 279 points
  Target (2.5 x ATR)    25,505
  Time stop             15 sessions
  Position size         5,000 / 279 = 17.9 -> 17 units
  Actual rupee risk     17 x 279 = Rs 4,743  (0.95R)
  Notional              17 x 25,040 = Rs 4,25,680
  Planned R:R           2.5 / 1.5 = 1.67

--- STATE ---
  Sleep last night      7.5 hrs
  Emotional state       Calm / neutral
  Last 3 trades         W, L, L
  Am I trading to
```

```
    recover a loss?     NO   (if YES → DO NOT TRADE)
  Conviction (1-5)      3    (it is a rule trade; conviction
                              is irrelevant and recorded only
                              to check whether it correlates
                              with outcomes - it usually does
                              not, which is the point)


TRADE JOURNAL - POST-TRADE ENTRY
================================
TRADE ID      : 2026-0147
EXIT DATE     : 2026-11-04   (10 sessions held)

--- EXECUTION ---
  Planned entry         25,040
  Actual entry          25,058      slippage  -18 pts
  Planned stop          24,761
  Actual exit           25,505       reason: TARGET
  Exit slippage                      -4 pts (filled 25,501)

--- RESULT ---
  Gross P&L             (25,501 - 25,058) x 17 = Rs  7,531
  Costs (all-in)                                 Rs    412
  Net P&L                                        Rs  7,119
  R-multiple            7,119 / 4,743           =    +1.50R
  Cost as % of gross                                 5.5%

--- RULE ADHERENCE ---
  Entered per rule?             YES
  Stop placed immediately?      YES
  Stop ever widened?            NO
  Size per formula?             YES
  Exited per rule?              YES
  Any deviation?                NONE
  PROCESS GRADE                 A

--- REVIEW ---
  What went right:
    "Followed the specification exactly. Did not interfere
     during a 4-session period where the trade was underwater
     by 0.6R."
  What went wrong:
    "Entry slippage of 18 points was larger than the 0.05%
     modelled (12.5 points). Three of the last five trades
     have had worse-than-modelled entry slippage. ACTION:
     re-measure slippage over the next 20 trades and update
     the cost model if the pattern holds."
  What I would do differently:
    "Nothing on this trade. The process worked."
```

```
  OUTCOME GRADE                Profitable
  PROCESS GRADE                A
  (Record BOTH. They are different things.)

--- TAGS FOR LATER ANALYSIS ---
  #pullback #uptrend #vix-low #target-hit #no-deviation
```

A machine-readable journal schema

JSON

```
{
  "trade_id": "2026-0147",
  "strategy": "index-pullback-continuation",
  "strategy_version": "0.3",
  "setup_tag": "PULLBACK-TO-20SMA-IN-UPTREND",
  "instrument": "INDEX-FUT",
  "direction": "long",
  "pre_trade": {
    "timestamp": "2026-10-21T09:42:00+05:30",
    "hypothesis": "uptrend intact, pullback below 20SMA, first up close",
    "invalidation": "close below 50SMA or below 200SMA",
    "rules_passed": ["E1","E2","E3","E4","E5","E6","F1","F2","F3"],
    "equity": 500000,
    "risk_frac": 0.01,
    "r_value_rupees": 5000,
    "planned_entry": 25040,
    "atr14": 186,
    "planned_stop": 24761,
    "planned_target": 25505,
    "time_stop_bars": 15,
    "position_units": 17,
    "actual_risk_rupees": 4743,
    "portfolio_heat_after": 0.02,
    "emotional_state": "neutral",
    "recovering_a_loss": false,
    "conviction_1_5": 3
  },
  "post_trade": {
    "exit_timestamp": "2026-11-04T09:16:00+05:30",
    "bars_held": 10,
    "actual_entry": 25058,
    "actual_exit": 25501,
    "exit_reason": "target",
    "gross_pnl": 7531,
    "costs": 412,
    "net_pnl": 7119,
    "r_multiple": 1.5,
    "entry_slippage_points": 18,
    "exit_slippage_points": 4,
    "rule_adherence": {
      "entered_per_rule": true,
      "stop_placed_immediately": true,
      "stop_widened": false,
      "size_per_formula": true,
      "exited_per_rule": true,
      "deviations": []
    },
    "process_grade": "A",
```

```
    "lessons": [
      "entry slippage exceeding model for 3 of last 5 trades",
      "action: re-measure over next 20 trades"
    ],
    "tags": ["pullback","uptrend","vix-low","target-hit","no-deviation"]
  }
}
```

What to compute from the journal

Analysis	Question it answers	Typical finding
Expectancy by setup tag	Which of my setups actually works?	Usually one or two tags carry all the profit and the rest are noise or negative. Cut the rest.
Expectancy by process grade	Does following my rules actually pay?	If A-graded trades have higher expectancy than C-graded, your edge is real and your discipline is the constraint.
Expectancy by day of week / time of day	Is there a time I should not trade?	Beware multiple testing here — with 5 days and 6 hour-buckets you have 30 comparisons and will find a "pattern" by chance.
Expectancy by emotional state	Does my state predict outcomes?	Frequently yes. Trades entered while "frustrated" or "excited" tend to be worse. This is a rule you can act on.
Expectancy after a loss / after a win	Am I revenge trading or over-confident?	If trades immediately after a loss have materially worse expectancy, institute a cooling-off rule.
Slippage versus model	Is my cost assumption still valid?	Drifts over time as liquidity changes. Must be re-measured, not assumed.
Deviation count per week	Is my discipline degrading?	The leading indicator of the big loss. Track it like a production error rate.
Distribution of R-multiples	Is the profit broad or concentrated?	If removing the top three trades turns the year negative, you do not have an edge yet.

Review cadence

1 **Daily (10 minutes, after the close)**

Complete post-trade entries for anything closed. Note any deviation. Do not analyse — just record. Analysis after a losing day is contaminated.

2. **Weekly (45 minutes, weekend)**
   Re-read every pre-trade entry from the week and ask whether the hypothesis was honest. Count deviations. Compute week's R total and running expectancy. Note anything to investigate.

3. **Monthly (2 hours)**
   Full metrics: expectancy, win rate, payoff ratio, profit factor, drawdown, cost ratio, expectancy by tag and by process grade. Compare against the backtest confidence bands.

4. **Quarterly (half a day)**
   Go/no-go decision on each strategy against the kill criteria. Re-measure the cost model. Review whether market regime has changed. Decide what to stop doing.

5. **Annually**
   Compare total net after-tax return against simply having bought a broad index fund with the same capital. Be honest about the hours spent. This is the only comparison that matters.

Trading psychology, concretely

Most writing on trading psychology is vague encouragement. This section treats psychological failures as *observable behaviours with measurable costs and specific countermeasures*, because that is the only version of the topic that is actionable.

> **The central reframe**
> "Fear" and "greed" are not causes. They are symptoms of a position that is too large. A 0.5%-risk position does not produce fear. A 6%-risk position produces fear in everyone, including professionals. When you notice an emotional problem, the first hypothesis should always be a sizing problem, not a character problem.

The named failure modes

Failure mode	Observable behaviour	Typical cost	Countermeasure
Revenge trading	A trade entered within minutes of a loss, outside the specification, at above-normal size	2–3× normal risk on a trade with no edge; often the largest single loss of the month	Hard rule: no new trade for 60 minutes after a loss. After three consecutive losses, stop for the day.
Overconfidence after a win streak	Size creeping up; rule checks skipped; "I have a feel for this market"	Guarantees your largest position coincides with your eventual worst outcome	Size is a formula, not a judgement. Review the formula after every win streak of 4+.
Loss aversion	Holding losers past the stop; "it will come back"; moving the stop	Converts −1R into −3R or worse; the single most expensive habit	Place the stop as a live order at entry. Never cancel it. If you cannot do this, you cannot trade.
Disposition effect	Cutting winners at +0.4R while holding losers to −2R	Destroys the payoff ratio, which destroys expectancy regardless of win rate	Pre-set the target. Journal every early exit and compute what it cost over 50 trades.
FOMO entry	Chasing a move already in progress, at a worse price, without a setup	Entry far from the logical stop, so either the stop is too wide or the size is too big	If the entry trigger has passed, the trade is gone. Missing trades is free; bad entries are not.
Analysis paralysis	Endless research; never placing the trade the system called for	You get the drawdowns (from the trades you did take) without the winners (from the ones you skipped)	Reduce to a binary checklist. If all boxes tick, the trade is taken. Discretion to skip is discretion to fail.
Boredom trading	Trades on quiet days with no signal, "just to be in the market"	Pure cost with zero edge; 10 a month is roughly ₹24,000 a year at retail sizes	Maintain a "no trade today" log entry. Count no-trade days as successful days.

Tilt	Rapid-fire trading, abandoning the plan entirely, size escalation	Account-ending. This is where most blow-ups actually occur.	The circuit-breaker rule below. Non-negotiable, pre-committed, automated if possible.
Confirmation seeking	Searching for opinions that agree with an open position	Delays the exit and reinforces the thesis past its invalidation	Write the invalidation condition in the pre-trade entry. Check it against price, not against opinion.
Attribution error	"I won because I'm good; I lost because the market is rigged"	Prevents all learning; you cannot improve what you do not own	The process grade. Grade the decision separately from the outcome, every time.

The effect of leverage on judgement

TEXT

```
WHY LEVERAGE DEGRADES DECISION-MAKING
═══════════════════════════════════════

Consider the same adverse 2% market move at three leverage
levels on a Rs 5,00,000 account.

Leverage   Notional      2% move    As % of    Felt as
                        against    account
─────────  ────────────  ─────────  ─────────  ──────────────
  1x       Rs 5,00,000   -10,000      -2.0%    "a normal day"
  3x       Rs 15,00,000  -30,000      -6.0%    "that hurt"
  5x       Rs 25,00,000  -50,000     -10.0%    "I need to act"
 10x       Rs 50,00,000 -1,00,000    -20.0%    panic

AT 1x you can think. At 10x you cannot, because:
  - The amygdala response to a 20% loss is not optional
  - You now face a margin call, which imposes a deadline
  - Every subsequent decision is made under time pressure
    and financial threat
  - The "correct" decision (hold, per the plan) and the
    "survival" decision (close everything) conflict

LEVERAGE DOES NOT JUST MULTIPLY P&L.
IT MULTIPLIES THE EMOTIONAL AMPLITUDE OF EVERY TICK, AND
EMOTIONAL AMPLITUDE IS INVERSELY RELATED TO DECISION QUALITY.

THE PRACTICAL TEST
  Find the position size at which you can walk away from the
  screen for two hours without anxiety. That is your maximum
  size. It is almost certainly smaller than what you are
  currently trading.

  If you cannot sleep with the position on, the position is
  too big. This is not a metaphor - it is a usable, reliable,
  calibrated instrument that you already own.
```

The circuit-breaker rule

TEXT

```
PRE-COMMITTED CIRCUIT BREAKERS
==============================
Write these down. Put them where you can see them. Automate
them if your broker allows it.

TRADE-LEVEL
  - Stop order placed at the same moment as the entry order.
    Not after. Not "in a minute". The same moment.
  - Stop is never widened. There is no exception clause.

DAILY
  - Maximum 3 losing trades, then stop for the day.
  - Maximum daily loss 3% of equity, then stop for the day.
  - Maximum 5 trades per day regardless of outcome.
  - 60-minute cooling-off after any loss before a new entry.

WEEKLY
  - Maximum weekly loss 6% of equity, then stop for the week.
  - If 2+ rule deviations occurred this week, stop for the
    following week and review.

MONTHLY
  - Maximum monthly loss 10%, then stop for the month and
    conduct a full review before resuming.

ACCOUNT-LEVEL
  - At -15% from peak: halve position size.
  - At -20% from peak: stop and conduct a full review.
  - At -25% from peak: return to paper trading until 30
    trades at the specified expectancy are demonstrated.

PERSONAL-STATE
  Do not trade when:
  - You slept under 6 hours
  - You are ill, angry, grieving, or under acute stress
  - You have had alcohol in the last 12 hours
  - You are in a financial emergency and "need" a win
  - You have a major deadline at work today
  - You just had an argument

THE ONLY RULE ABOUT THESE RULES
  They must be written BEFORE you need them, because the
  moment you need them is the moment you will be least
  capable of writing them. A rule invented during a drawdown
  is a rationalisation, not a rule.
```

If you break a circuit breaker once, you will break it again

Treat the first violation as a serious incident. Log it. Analyse it like a production outage: what was the trigger, what was the state, what control failed, what control would have prevented it. Traders who normalise rule violations do not gradually deteriorate — they continue apparently fine until one violation coincides with a large adverse move, and then the account is gone.

Routine and process

Discretion fails under pressure; routine does not. The purpose of a routine is to move as many decisions as possible out of the live session, where your judgement is worst, into prepared time, where it is best.

The pre-market routine

1. **Personal state check (2 min)**
   Sleep, health, stress, deadlines, emotional state. If any flag is raised, today is a no-trade day. Log it as such.
2. **Account state (3 min)**
   Current equity, open positions, current portfolio heat, distance from the drawdown circuit breakers, available margin, any pending corporate actions or expiries.
3. **Calendar check (3 min)**
   Scheduled events today and in the next three sessions — policy decisions, major data releases, results for anything you hold, expiry dates, and any no-trade windows your specification defines.
4. **Scan for signals (10–15 min)**
   Run the specification over the universe. Produce a list. Do not look at anything outside the list.
5. **Pre-compute every trade (10 min)**
   For each candidate: entry, stop, target, size, rupee risk, resulting heat. Write the pre-trade journal entry now, before the session opens, while you are calm.
6. **Write the no-trade conditions (1 min)**
   "I will not trade if X." Having it written removes the mid-session argument.
7. **Close the news**

Social media, television, forums. You have your list. Nothing you read now will improve it; it can only cause deviation.

In-session rules

- [ ] Execute only trades from the pre-market list. No additions. Ever.
- [ ] Place the stop order at the same moment as the entry order.
- [ ] No position size changes to an open trade unless the specification defines scaling.
- [ ] No chart-watching between signals. Set alerts and close the terminal.
- [ ] No P&L watching. Watching P&L causes exits; watching the plan causes execution.
- [ ] If you feel the urge to deviate, write down the proposed deviation and the reason before acting. Most urges do not survive being written down.
- [ ] Honour the daily circuit breakers immediately, without negotiation.
- [ ] Avoid the first and last 15 minutes unless the strategy explicitly requires them.

Post-market, weekly and monthly

Cadence	Duration	Content	The thing people skip
Post-market	10 min	Complete post-trade entries, log deviations, note the day's R total, log a no-trade day if applicable	Logging no-trade days — which are the days you avoided a loss
Weekly	45 min	Re-read pre-trade hypotheses, count deviations, compute weekly R, review open positions against their invalidation conditions	Re-reading your own pre-trade reasoning with fresh eyes
Monthly	2 hrs	Full metrics, expectancy by tag and by process grade, cost ratio, slippage versus model, drawdown status	Comparing live statistics against the backtest confidence bands
Quarterly	Half day	Kill-criteria check on every strategy, cost model refresh, regime assessment, decide what to stop	Actually stopping something
Annual	Half day	After-tax return versus a passive benchmark with the same capital; hours spent; effective hourly rate	The effective hourly rate, which is usually sobering

When not to trade at all

NO SIGNAL

The specification produced no candidates. This is the most common correct outcome. A trader who trades every day has a specification with no filter, which is to say no strategy.

REGIME FILTER OFF

Volatility outside the tested range, or the trend filter inactive. The strategy was never tested in this regime; running it here is live experimentation with real money.

EVENT WINDOWS

Policy decisions, budgets, results for a position you hold. Unless your strategy is specifically an event strategy, these are periods of unmodelled risk.

PERSONAL STATE

Under-slept, ill, stressed, angry, recently argued, drinking last night, major work deadline. The cost of not trading is zero; the cost of trading badly is not.

CIRCUIT BREAKER ACTIVE

Daily, weekly or monthly loss limit hit. The rule exists precisely for the moment when you want to override it.

NEEDING MONEY

If you need this month's trading to cover an expense, your decisions are already compromised. Trading capital must be genuinely discretionary.

Algorithmic and quantitative trading for retail in India

As a software engineer you have a genuine advantage here — you can build, test and automate. That advantage is real but narrower than it feels, and the regulatory framework for retail algorithmic trading in India has been tightened.

⚠ **The regulatory position has changed and is still evolving**
SEBI has issued circulars and consultation papers on algorithmic trading by retail investors, covering broker responsibilities, the registration and approval of algorithms, API access, unique identifiers for algo orders, order-per-second thresholds, and the treatment of third-party algo providers. The framework has been revised and implementation dates extended. **As at late 2026, verify the current position at sebi.gov.in** and with your broker before automating anything.

What the framework broadly requires

Area	Direction of the rules (verify current detail)
Broker responsibility	Brokers are responsible for the algos routed through their systems, including those built by clients via APIs.
Algo registration / tagging	Algorithmic orders are expected to be identifiable, with unique identifiers so that exchanges can trace orders to a specific algo.
Order rate thresholds	Above a specified order-per-second rate, additional requirements (including exchange approval of the algo) apply. Below it, lighter treatment for self-developed algos by individual investors.
Third-party algo providers	Brought within the framework through broker empanelment; the marketing of algos with performance claims is restricted.
Self-developed algos	Individual investors developing their own algos have a defined path, subject to broker processes and the order-rate thresholds.
Prohibited claims	Advertising assured or indicative returns from an algorithm is not permitted.

✕ **Be very wary of "algo trading" products sold to retail**
A large industry sells "proven algos" with claimed returns to retail investors in India. SEBI has acted against assured-return claims in this space. If a product promises a return, shows an unaudited equity curve, or asks for a profit share without a clear regulatory status, treat it as a scam until proven otherwise. A genuine algorithmic edge is not sold for ₹5,000 a month to strangers.

What an individual can and cannot compete on

Dimension	Professional firm	You	Verdict
Latency	Co-located servers, kernel bypass, FPGA, microsecond round trips	Home broadband, broker API, hundreds of milliseconds	**Not competitive — do not build anything latency-sensitive**
Data	Full tick depth, historical order book, alternative data, vendor feeds	End-of-day or delayed intraday, limited history, often inconsistent	**Significant disadvantage**
Costs	Negotiated, exchange membership, no brokerage spread	Retail tariff plus GST plus STT	**Significant disadvantage**
Capital	Large, diversified across strategies	Small, concentrated	**Disadvantage, with one upside — see below**
Capacity constraints	Cannot trade small, illiquid opportunities profitably	**Can** — small size is the one structural advantage you have	**Advantage**
Mandate constraints	Must stay invested; benchmark-relative; quarterly reporting pressure	Can hold cash indefinitely; no reporting; no career risk	**Advantage**
Patience	Limited by investor and employer expectations	Unlimited, if you have the temperament	**Advantage**
Research quality	PhD teams, peer review, risk oversight	You, alone, with confirmation bias	**Disadvantage — mitigate with rigour**

✦ **The only quadrant where a retail quant has a real chance**

Low-frequency, capacity-constrained, patience-requiring strategies in corners too small for institutional capital. Nothing latency-sensitive. Nothing requiring exotic data. Nothing that needs to be traded a thousand times a day. If your algorithmic idea requires speed, better data or lower costs than you have, it is not an idea — it is a description of someone else's business.

Infrastructure a retail quant actually needs

TEXT

```
A MINIMAL RETAIL QUANT STACK
============================

DATA
  - Historical daily OHLCV with corporate-action adjustment
  - An exchange trading calendar (NOT a generic business-day
    calendar - Indian market holidays and special sessions
    will break naive date logic)
  - Point-in-time index constituents if you trade stocks
  - A second data source for cross-validation
  STORE IT LOCALLY. Vendor APIs change, go down, and
  silently revise history.

RESEARCH
  - Python, pandas, numpy
  - A backtest engine you WROTE YOURSELF at least once, so
    you understand where the lies hide. Then use a library.
  - Version control for the strategy code AND the parameters
  - A results database: every experiment, every parameter
    set, every outcome - including the failures. Without
    this you cannot assess your own multiple-testing burden.

EXECUTION
  - Broker API with documented rate limits
  - An order management layer with: idempotent order
    placement, reconciliation against broker state on
    restart, a kill switch, and hard limits that the
    strategy code cannot override
  - Logging of every decision, every order, every fill,
    with timestamps

MONITORING
  - Position and P&L reconciliation against the broker,
    at least daily
  - Alerting on: disconnection, rejected orders, position
    mismatch, drawdown thresholds, order-rate anomalies
  - A DEAD MAN'S SWITCH: if the system cannot confirm its
    own health, it flattens positions and stops

THE ENGINEERING RULES THAT MATTER MOST
  1. The kill switch must be OUTSIDE the strategy code and
     must work even if the strategy is in a bad state.
  2. Risk limits are enforced at the order layer, not in
     the strategy. The strategy cannot be trusted.
  3. Reconcile against broker state on every restart.
     NEVER assume your in-memory position is correct.
  4. Fail closed. On any uncertainty, place no orders.
  5. Paper trade the FULL stack for months before live
     capital. Most failures are infrastructure failures,
```

```
   not strategy failures.
6. Log everything. You will need it for the post-mortem
   and possibly for a dispute with the broker.
```

PYTHON

```python
"""
Risk gate enforced OUTSIDE strategy code.
Illustrative skeleton. Not production code, not a recommendation.
"""

from dataclasses import dataclass, field
from datetime import date


@dataclass
class RiskLimits:
    max_risk_per_trade_frac: float = 0.01
    max_portfolio_heat_frac: float = 0.05
    max_positions: int = 5
    max_daily_loss_frac: float = 0.03
    max_orders_per_day: int = 20
    max_notional_frac: float = 0.60


@dataclass
class RiskState:
    equity: float
    open_risk_rupees: float = 0.0
    open_positions: int = 0
    realised_pnl_today: float = 0.0
    orders_today: int = 0
    open_notional: float = 0.0
    halted: bool = False
    halt_reasons: list = field(default_factory=list)


class RiskGate:
    """Every order passes through here. The strategy cannot
    bypass it. If any check fails, the order is rejected and
    the reason is logged."""

    def __init__(self, limits: RiskLimits):
        self.limits = limits

    def check(self, state: RiskState, order_risk_rupees: float,
              order_notional: float):
        reasons = []
        L = self.limits

        if state.halted:
            reasons.append("system halted: " + ",".join(state.halt_reasons))

        if order_risk_rupees > state.equity * L.max_risk_per_trade_frac:
```

```
            reasons.append("per-trade risk exceeds limit")

        new_heat = (state.open_risk_rupees + order_risk_rupees) / state.equity
        if new_heat > L.max_portfolio_heat_frac:
            reasons.append(f"portfolio heat {new_heat:.3f} exceeds limit")

        if state.open_positions ≥ L.max_positions:
            reasons.append("max concurrent positions reached")

        daily_loss_frac = -state.realised_pnl_today / state.equity
        if daily_loss_frac ≥ L.max_daily_loss_frac:
            reasons.append("daily loss circuit breaker tripped")

        if state.orders_today ≥ L.max_orders_per_day:
            reasons.append("daily order count limit reached")

        new_notional = (state.open_notional + order_notional) / state.equity
        if new_notional > L.max_notional_frac:
            reasons.append("gross notional exposure limit")

        return {"allowed": len(reasons) == 0, "reasons": reasons}


def reconcile(local_positions, broker_positions):
    """NEVER trust in-memory state after a restart.
    Any mismatch halts the system - it does not 'fix itself'."""
    mismatches = []
    keys = set(local_positions) | set(broker_positions)
    for k in keys:
        a = local_positions.get(k, 0)
        b = broker_positions.get(k, 0)
        if a ≠ b:
            mismatches.append({"symbol": k, "local": a, "broker": b})
    return {"ok": not mismatches, "mismatches": mismatches}
```

Dabba trading — do not

> ✕ **Dabba (bucket) trading is illegal**
>
> Dabba trading is off-exchange betting on the prices of listed securities, operated by unregistered intermediaries. Orders never reach the exchange. It is marketed on lower costs, no STT, higher leverage and no KYC. The reality: you have no exchange protection, no clearing corporation guarantee, no investor grievance mechanism, no legal recourse when the operator refuses to pay, and you are participating in an illegal activity that has attracted enforcement action and tax proceedings. If someone offers you trading with no STT and 50x leverage, you are being offered a crime with a high probability of being defrauded.

Scams, tip services and the "profitable system" industry

A large, profitable industry exists around selling trading signals, courses and "algos" to retail participants. Understanding its business model tells you everything about whether its products work.

The arithmetic of selling signals

TEXT

WHY SELLING TIPS IS MORE PROFITABLE THAN TRADING THEM

A tip service with 2,000 subscribers at Rs 3,000 per month:
  Monthly revenue = 2,000 x 3,000 = Rs 60,00,000
  Annual revenue = Rs 7,20,00,000 (seven crore twenty lakh)

  Costs: a Telegram channel, some staff, marketing. Low.
  Capital at risk: ZERO.
  Downside: zero, beyond reputational and regulatory risk.

To earn Rs 7.2 crore a year by TRADING at a (very good) 25%
annual return, you would need Rs 28.8 crore of capital,
and you would risk it every day.

THE INCENTIVE IS UNAMBIGUOUS. If you genuinely had a method
producing reliable returns, the rational action is to raise
capital and trade it, not to sell it for Rs 3,000 a month to
strangers who will dilute its capacity.

THE "FREE CALLS" FUNNEL

Stage 1: free Telegram channel, thousands of members
Stage 2: post many "calls" daily
Stage 3: post screenshots of the winners only
Stage 4: convert the impressed to a paid "VIP" group
Stage 5: in the VIP group, repeat with bigger claims
Stage 6: upsell a "mentorship" at Rs 50,000–2,00,000

THE SPLIT-SIGNAL TRICK

Take 8,000 free subscribers. Split into two groups of 4,000.
  Group A: "Index will go UP this week"
  Group B: "Index will go DOWN this week"

One group is right. Keep them; discard the others.
  Split the 4,000 winners again: 2,000 / 2,000
  One group is right again. Keep them.
  Repeat: 1,000 → 500 → 250

After five rounds, 250 people have seen you be RIGHT FIVE
TIMES IN A ROW. From their perspective you are clairvoyant.
Convert them to a paid group at Rs 10,000 a month.
Revenue: Rs 25,00,000 a month from a trick requiring zero
market knowledge.

This is not theoretical. Variants of it are run constantly.

THE SCREENSHOT PROBLEM

```
──────────────────────
A P&L screenshot proves nothing:
  - It is trivially edited; browser developer tools take
    about four seconds
  - A genuine Rs 4 lakh profit screenshot is fully compatible
    with a Rs 12 lakh year-to-date loss
  - It shows one position, not an account
  - It shows a day, not an equity curve
  - A demo/paper account looks identical to a real one

THE ONLY EVIDENCE THAT WOULD MEAN ANYTHING
  A complete, continuous, independently verified equity curve
  covering several years and multiple market regimes, with
  drawdowns, net of all costs and taxes, with the capital base
  disclosed. Essentially nobody in retail social media provides
  this, and the reason is not modesty.
```

How to verify a SEBI registration

1. **Ask for the registration number**
   A SEBI-registered Research Analyst (RA) or Investment Adviser (RIA) has a registration number in a defined format and is required to disclose it. Refusal to provide it is a complete answer.

2. **Check it on sebi.gov.in**
   SEBI publishes searchable lists of registered intermediaries. Verify the number, the name, the entity type and that the registration is current — not lapsed, suspended or cancelled.

3. **Check the category matches the service**
   A Research Analyst may publish research. An Investment Adviser may give personalised advice for a fee. Neither may guarantee returns. Someone registered as one and acting as the other is operating outside their registration.

4. **Check for enforcement orders**
   SEBI publishes orders and settlement records. Search the name and the entity.

5. **Check the fee structure**
   SEBI prescribes constraints on how registered advisers charge, including fee caps and the prohibition on handling client funds. An adviser asking for a profit share, access to your trading account, or direct transfers to a personal account is a serious warning sign.

6. **Never give trading-account access**
   No legitimate adviser needs your broker credentials. Handing them over means you have no recourse and possibly no insurance.

Red flag	Why it is disqualifying
Guaranteed or assured returns	Prohibited and impossible. Markets do not provide guarantees; anyone claiming one is lying.
"90% accuracy" or similar	Accuracy without payoff ratio is meaningless, and the number is unverifiable. Also a classic marketing claim rather than a statistic.
Urgency and scarcity	"Only 10 seats left", "offer closes tonight". Pressure tactics exist to prevent due diligence.
Profit sharing	Creates an incentive for maximum risk — the provider gets the upside and you keep all the downside.
Asking for account access or funds	Registered advisers may not handle client funds in this way. This is how money disappears.
No registration number, or an unverifiable one	Giving investment advice for consideration without registration contravenes SEBI regulations.
Testimonials and lifestyle marketing	Cars, watches, hotel lobbies. The income is from you, not from the market.
"No STT, higher leverage, no KYC"	Dabba trading. Illegal, with total counterparty risk.
Pressure to recruit others	A referral or multi-level structure means the revenue model is recruitment, not performance.
Deleting losing calls	If the channel's history is edited, the track record is fiction.

⚠ **A note on finfluencers**

SEBI has moved to restrict associations between registered intermediaries and unregistered persons who make performance claims, and has acted against unregistered advisory activity. The practical rule for you is simple: if someone is telling you what to buy and is not a registered RA or RIA with a verifiable number, they are not permitted to do so for consideration, and you have no recourse when it goes wrong. Verify the current position at sebi.gov.in.

The honest conclusion

Everything above assumed you were going to trade. This section asks whether you should, and tries to answer it without flattery.

Who should not trade

- ☐ Anyone without an emergency fund, insurance, and a funded long-term investment plan already running.
- ☐ Anyone trading with borrowed money, or with money that has a purpose.
- ☐ Anyone who cannot give the market consistent, sustainable time during session hours for the style they have chosen.
- ☐ Anyone who cannot write down a falsifiable hypothesis with an economic rationale.
- ☐ Anyone who has not tested a specification over 100+ trades with realistic cost modelling.
- ☐ Anyone who needs trading income to meet expenses.
- ☐ Anyone who has increased size to recover a loss, more than once.
- ☐ Anyone whose trading affects their sleep, their work, or their relationships.
- ☐ Anyone following tips, signals or paid "calls".
- ☐ Anyone who cannot state their expectancy, their maximum drawdown and their cost ratio from their own records.

Who might reasonably trade

- ☐ Someone with a secure income, a fully funded long-term portfolio, and genuinely discretionary risk capital.
- ☐ Someone who treats it as a serious, measured, loss-making-until-proven-otherwise activity rather than an income source.
- ☐ Someone with the engineering discipline to specify, test, instrument and review — and the honesty to kill a strategy that fails its criteria.

- [ ] Someone operating in a low-frequency, capacity-constrained niche where small size is an advantage rather than a handicap.
- [ ] Someone who finds the research genuinely interesting independent of the money, because for a long time there will be no money.
- [ ] Someone who has written down, in advance, the drawdown at which they will stop — and who will honour it.

The opportunity cost, computed

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WHAT ELSE COULD THOSE HOURS HAVE DONE?
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Assume 10 hours a week on trading: research, execution,
journaling, review. That is 500 hours a year.

ALTERNATIVE A — CAREER
  500 hours toward a specialism, a certification, open-source
  contribution, or a visible body of work.
  A plausible effect on a software engineer's compensation
  over 3 years: a 20–40% increase in total compensation.
  On a Rs 30 lakh package, a 25% increase is Rs 7.5 lakh
  PER YEAR, COMPOUNDING, RECURRING, with no capital at risk
  and no drawdowns.

ALTERNATIVE B — SIDE INCOME
  500 hours of freelance or consulting work at a modest
  Rs 2,500 per hour = Rs 12,50,000 a year, with a
  100% win rate and zero risk of ruin.

ALTERNATIVE C — TRADING
  500 hours, Rs 5,00,000 of capital, and roughly a 10%
  probability of a positive outcome based on the SEBI data.
  Expected value: negative.
  Best realistic case for a disciplined participant with a
  genuine small edge: perhaps Rs 50,000–1,00,000 a year,
  taxed at the marginal rate, i.e. Rs 70–140 per hour
  before accounting for the risk taken.

THE COMPARISON IS NOT CLOSE, AND IT IS THE COMPARISON ALMOST
NOBODY RUNS.

For an Indian software engineer specifically, the highest
expected-value financial activity is almost always:
  1. Increase income (career, skills, specialisation)
  2. Increase savings rate
  3. Invest the surplus systematically in low-cost index funds
  4. Do not touch it

Trading is, at best, a fourth priority and an expensive hobby.
It becomes defensible only when the first three are complete
and the capital involved is genuinely surplus.
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✓ **The most useful thing this guide can tell you**

The skills in it — thinking in expectancy, sizing for survival, testing honestly, separating decision quality from outcome, pre-committing to rules, journaling and reviewing — are enormously valuable. They are also transferable to every important decision you will make, in engineering and in life. You can acquire all of them from a small paper-trading programme and a journal, at zero financial risk. Acquiring them is the win. Trading with real money is optional, and for most people it is the part that should be skipped.

Frequently Asked Questions

What percentage of retail traders in India actually make money?

SEBI's study published in September 2024, covering FY 2021–22 to FY 2023–24, found roughly 93% of individual F&O traders made losses, with aggregate individual net losses in the region of ₹1.8 lakh crore. A 2025 follow-up covering FY 2024–25 found roughly 91% of individual index-option traders losing. Studies of cash-segment intraday trading show a similar majority losing, with the loss rate rising with trade frequency. Verify the current figures at sebi.gov.in — the methodology and periods change, but the direction has been consistent across every version.

Can I realistically trade while working full time as a software engineer?

Intraday and scalping: no. Not "difficult" — structurally incompatible, because those styles require continuous attention during exactly the hours you are being paid to do something else, and they are also the most cost-sensitive styles. Swing and positional trading on a daily timeframe are compatible: signals are evaluated after the close, orders are placed before the open, and the time demand is 30–60 minutes a day plus a weekly review. If your plan requires watching charts during work hours, the plan is already broken and you will do both things badly.

How much capital do I need to start?

Enough that fixed costs do not dominate and that a single position can be sized at 1–2% risk. Below roughly ₹2 lakh, flat per-order brokerage and minimum lot sizes make responsible position sizing arithmetically impossible — one position becomes 15–30% of the account. More importantly, the capital must be genuinely discretionary: money whose total loss changes nothing material in your life. For most people that number is smaller than their enthusiasm, and the honest answer is to paper trade until the capital exists.

Is technical analysis useful or is it astrology?

Neither extreme is right. Some concepts have genuine content — volatility measures like ATR for stop placement and sizing, liquidity and volume for execution quality, and simple trend filters as regime indicators. Much of the rest — named chart patterns, Fibonacci levels, elaborate indicator combinations — has weak and regime-dependent evidence, is visible to millions of participants

simultaneously, and is extremely prone to overfitting when tested. The practical position: use a minimal number of simple, economically motivated indicators, test them honestly out of sample, and be suspicious of anything that requires you to interpret a chart subjectively, because subjectivity cannot be backtested.

What is a realistic return for a good retail trader?

The question is badly posed, because the expected value for an untested retail trader is negative. If you have demonstrated a positive expectancy out of sample over 200+ trades, and you size at 1% risk with a +0.20R expectancy and 120 trades a year, the arithmetic gives roughly 24% a year gross before tax — with drawdowns likely to reach 20–30% along the way, and with a wide distribution around that figure. After the marginal tax rate on business income, that becomes roughly 16–17% net. Compare that honestly to a passive index portfolio requiring two hours a year. The comparison is closer than most traders want it to be, and that is for the *successful* case.

How do I know if my strategy actually has an edge?

You cannot know with certainty; you can only accumulate evidence. The minimum bar: a written specification with no hidden discretion, 200+ trades including a genuinely held-out out-of-sample period, realistic cost and slippage modelling, parameter sensitivity showing a broad plateau rather than a sharp peak, performance across at least two distinct market regimes, and a Monte Carlo reshuffle showing a drawdown you could survive. Even then, deflate your measured expectancy substantially to account for the multiple testing you did not count. Fewer than 100 trades tells you essentially nothing, which is why most retail conviction is formed on noise.

What is the most common reason retail traders lose?

In order of magnitude: (1) no edge to begin with, compounded by the near-zero-sum structure of derivatives; (2) transaction costs, which scale linearly with frequency and routinely exceed 30–40% of a small account per year; (3) position sizing that is far too large, making normal variance fatal; (4) failure to honour stops, converting −1R losses into −3R losses; and (5) behavioural escalation — revenge trading, averaging down, and size increases after wins. Note that only the first is about market knowledge. Four of the five are about process and discipline, which are fully within your control.

Should I use a stop loss on every trade?

You need a defined exit on every trade — that is non-negotiable. Whether it is a resting stop order depends on the instrument and the strategy. For liquid instruments and intraday or swing strategies, place the order. For positional strategies in volatile instruments, a stop placed inside the normal noise range will take you out repeatedly at a cost; a wider stop with a smaller position is usually better. Understand that no stop protects against an overnight gap or a lower-circuit lock, so your position sizing must survive an outcome roughly three times worse than the stop implies. A "mental stop" is, for most beginners, not a stop at all.

How much should I risk per trade?

One to two percent of trading capital, with a hard cap on total portfolio heat of around 5%. The Monte Carlo evidence is stark: with a genuinely positive-expectancy strategy, risking 2% per trade gives roughly a 14% chance of a 50% drawdown over 500 trades, while risking 5% gives roughly a 68% chance. Risk of ruin is sharply non-linear in position size. Professional practitioners typically operate at an eighth to a quarter of full Kelly, which lands in the same 1–4% range. If your sizing exceeds half Kelly on your own (probably optimistic) estimates, you are over-betting.

What drawdown should I expect?

Larger and longer than your backtest suggests. Take your backtested maximum drawdown, compute the Monte Carlo reshuffled 5th-percentile drawdown, take the larger, and multiply by 1.5 — that is your planning number. For a trend-following strategy with a 35% win rate, expect losing streaks of 14 or more trades over a few hundred trades, which means weeks of consecutive losses. The *duration* of a flat period is usually harder to endure than the depth of the decline. Plan for a full year with no progress and decide now whether you could tolerate that.

How are trading profits taxed in India?

Broadly: intraday equity is treated as **speculative business income**; exchange-traded F&O is **non-speculative business income**; delivery-based equity is capital gains unless the facts indicate a business. Business income is taxed at slab rates, which for a well-paid engineer means the top marginal rate — materially worse than the concessional capital-gains treatment on long-held equity. Business treatment does allow expense deduction (brokerage, data, equipment depreciation, professional fees). Turnover for F&O has a specific definition and a tax audit may be required above certain thresholds. Losses are only carried forward if the return is filed on time. **As at FY 2026–27, verify at incometax.gov.in** — note the Income-tax Act 2025 took effect from 1 April 2026 — and engage a chartered accountant.

Is algorithmic trading allowed for retail investors in India?

Yes, within a framework. SEBI has issued rules covering broker responsibility for algos routed through their systems, unique identification of algo orders, order-per-second thresholds above which exchange approval is required, the treatment of third-party algo providers via broker empanelment, and a defined path for self-developed algos by individual investors. Advertising assured or indicative returns from an algorithm is not permitted. The framework has been revised and implementation extended more than once — **verify the current position at sebi.gov.in** and with your broker before automating anything. Separately: be extremely sceptical of "proven algo" products sold to retail with performance claims.

How do I verify whether an advisor or tip service is legitimate?

Ask for their SEBI Research Analyst or Investment Adviser registration number and verify it directly in SEBI's published lists at sebi.gov.in — check the name, the category, and that the registration is current. Check for enforcement orders against them. Confirm the fee structure complies with SEBI's constraints: no guaranteed returns, no profit sharing, no handling of your funds, no access to your trading account. If they refuse to supply a number, show only edited screenshots, use urgency tactics,

or offer "no STT, high leverage" arrangements (which is dabba trading, and illegal), stop immediately. The economics are also informative: someone with a genuine edge raises capital and trades it rather than selling signals for ₹3,000 a month.

Should I quit my job to trade full time?

No. The base rates alone make this indefensible, but there is a second reason that is specific and practical: trading with capital you need to live on destroys the decision-making that trading requires. Every position becomes a rent payment. You will take trades you should skip, hold losers you should cut, and size up to meet a monthly target that the market has no obligation to supply. People who successfully trade full time have typically done so for years part time first, with a demonstrated and audited edge, and with living expenses covered from capital or another source for a long runway. If you are asking the question, the answer is no.

What should I do instead, if I conclude trading is not for me?

Direct your effort to the three things with far better expected value: increase your income (skills, specialisation, visible work — a 25% compensation increase on a ₹30 lakh package is ₹7.5 lakh a year, recurring, with no capital at risk); increase your savings rate; and invest the surplus systematically in low-cost diversified index funds with an asset allocation suited to your horizon. Automate it and leave it alone. Then keep the analytical skills from this guide — expectancy thinking, base rates, honest testing, separating decision quality from outcome — and apply them to engineering and career decisions, where the odds are overwhelmingly in your favour.

Trading-readiness checklist and a 90-day paper-trading programme

Two closing tools. The first is an honest gate. The second is a structured programme that costs nothing and will tell you, within a quarter, whether you should proceed.

Readiness checklist

GATE 1 — FINANCIAL PRECONDITIONS (EVERY ITEM MUST BE TRUE)

- ☐ Emergency fund of 6–12 months of expenses, fully funded and untouched.
- ☐ Health insurance in place; term life cover if anyone depends on you.
- ☐ No high-interest debt — no credit card balances, no personal loans.
- ☐ A long-term investment plan running on automatic, funded monthly.

- [ ] Trading capital is genuinely surplus; its total loss changes nothing material.
- [ ] No borrowed money in the trading account; no long-term portfolio pledged as margin.
- [ ] A separate account for trading, with its own accounting.

GATE 2 — KNOWLEDGE PRECONDITIONS

- [ ] I can compute expectancy in R, and I know why win rate alone is meaningless.
- [ ] I can compute position size from equity, risk percentage and stop distance.
- [ ] I know my round-trip cost as a percentage of capital at risk, for the instruments I trade.
- [ ] I understand gap risk and circuit limits and have sized for a 3R adverse outcome.
- [ ] I understand the difference between in-sample and out-of-sample, and why 85 trades with 15 parameters proves nothing.
- [ ] I know the tax treatment of what I intend to trade and the record-keeping it requires.
- [ ] I can explain, in one sentence, who is on the other side of my trade and why they are wrong.

GATE 3 — PROCESS PRECONDITIONS

- [ ] A written strategy specification with hypothesis, universe, timeframe, entry, exit, stop, filters, sizing, cost model and kill criteria.
- [ ] A backtest over 200+ trades with realistic costs, with a held-out out-of-sample period touched once.
- [ ] Parameter sensitivity shows a plateau, not a peak.
- [ ] A Monte Carlo drawdown estimate I could financially and emotionally survive.
- [ ] A journal template ready, with pre-trade and post-trade sections and a process grade.
- [ ] Written circuit breakers: per-trade, daily, weekly, monthly and account-level.
- [ ] A pre-committed drawdown level at which I stop entirely.
- [ ] A pre-committed date at which I compare results against a passive benchmark and decide whether to continue.

✕ **Any unchecked box in Gate 1 is disqualifying**

Gates 2 and 3 can be completed with study and work. Gate 1 cannot be worked around. If the emergency fund is not there, or the debt is not cleared, or the capital is not genuinely surplus, no amount of skill will make the activity appropriate.

The 90-day paper-trading programme

Zero financial risk. The objective is not to "practise making money" — it is to find out whether you have an edge and whether you can execute a process, before either question costs you anything.

Week	Objective	Deliverable	Pass criterion
1	Write the strategy specification	A complete spec: hypothesis with economic rationale, universe, timeframe, entry, exit, stop, filters, sizing, cost model, metrics, kill criteria	A competent stranger could reproduce your trade list from it
2	Build the cost model	A spreadsheet or function computing full round-trip cost for your instruments, with current published rates and a measured slippage assumption	You can state cost as a % of capital at risk for every trade size you will use
3–4	Backtest in-sample	Trade list, equity curve, full metrics, parameter sensitivity surface	Expectancy > +0.20R after costs on at least 100 in-sample trades, with a parameter plateau
5	Out-of-sample test	Results on the held-out period, touched exactly once, with no parameter changes	Expectancy > +0.10R after costs. If not, stop: the hypothesis is rejected
6	Monte Carlo and stress tests	Reshuffled drawdown distribution; results with doubled costs, with the top 5% of trades removed, and with entries shifted by one bar	Survives all four. 5th-percentile drawdown is one you could live through
7	Set up the journal and circuit breakers	Journal template in use; circuit breakers written and visible; a daily routine documented	You have run one full day of the routine end to end
8–10	Paper trade, phase 1	Every signal taken, logged pre-trade and post-trade, with honest fills (assume the worse side of the spread)	Zero rule deviations. Not "few". Zero. Expectancy is secondary this phase.
11	First formal review	Expectancy, win rate, payoff ratio, cost ratio, deviation count, expectancy by process grade	Live-simulated statistics within the backtest confidence bands

12–13	Paper trade, phase 2	Continue, with a deliberate stress test: take a scheduled break, trade after a losing streak, trade on a day you feel off	Rules honoured under stress; no-trade days correctly logged
14–15	Deviation analysis	Every deviation documented as an incident: trigger, state, control that failed, control that would prevent it	Fewer than 2 deviations across the whole programme, with written controls for each
16–17	Paper trade, phase 3	Continue to 60+ simulated trades total	Expectancy still positive after costs; drawdown within expectations
18	Go / no-go decision	A written decision document with all metrics, against pre-committed criteria	See the decision rule below

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THE 90-DAY GO / NO-GO DECISION RULE
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Write this down in week 1, before you have any results.

GO (proceed to live trading at MINIMUM size) requires ALL of:
  [ ] Out-of-sample backtest expectancy  > +0.10R after costs
  [ ] Paper-trading expectancy           > +0.10R after costs
  [ ] Simulated trades                   ≥ 60
  [ ] Rule deviations                    ≤ 2 across 90 days
  [ ] Monte Carlo 5th-pctile drawdown    survivable
  [ ] Cost ratio (costs / gross profit)  < 40%
  [ ] Gate 1 financial preconditions     all true
  [ ] Written circuit breakers           in place and tested
  [ ] A written stop-trading drawdown    committed

IF ALL NINE ARE TRUE
  Go live at HALF your intended risk per trade (0.5%, not 1%)
  for the first 50 live trades. Live results are systematically
  worse than paper results - slippage is real, fills are worse,
  and emotion is present for the first time.
  Re-review after 50 live trades before increasing size.

IF ANY ARE FALSE
  Do not go live. Choose one:
   (a) Fix the specific failure and extend the programme by
       30 days - ONCE. Not repeatedly; repeated extension is
       overfitting your own process.
   (b) Reject the hypothesis and either develop a new one or
       conclude that trading is not for you.

  Option (b) after 90 days of honest work is a SUCCESS, not a
  failure. You will have learned the most valuable lesson
  available in this domain at a cost of zero rupees, which is
  precisely the opposite of how most people learn it.

THE HONEST PRIOR
  Most specifications fail at week 5 (out-of-sample) or at
  week 14 (deviation count). Both failures are informative.
  The out-of-sample failure tells you the edge was never
  there. The deviation failure tells you that even with an
  edge, you would not have executed it.
  Each costs nothing to discover now, and a great deal to
  discover with money on the table.
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⚠ **One last honest note about paper trading**

Paper results overstate live results, consistently and materially. There is no slippage you did not choose, no partial fill, no rejected order, no margin call, no broker outage, and above all no fear. Model costs aggressively, assume you get the worse side of every spread, and treat a marginally positive paper expectancy as equivalent to a negative live one. If your paper expectancy is +0.12R, plan as though it is zero.

✕ **Closing note**

This guide has given you the maths, the process and the standards of serious trading. It has deliberately not given you a strategy, because any edge that can be written down and distributed stops being an edge. The evidence from SEBI's own participant data is that the overwhelming majority of Indian individuals who trade actively lose money, and that the losses are very large in aggregate. Nothing in this guide changes that base rate; it only changes whether you find out cheaply or expensively. If you work through the 90-day programme honestly and conclude that you do not have an edge, you have achieved the best realistically available outcome. Every figure, rate, charge, tax treatment and regulation above is illustrative and dated to late 2026 / FY 2026–27; verify current details at **nseindia.com**, **sebi.gov.in** and **incometax.gov.in**, and consult a SEBI-registered Investment Adviser and a chartered accountant for personal advice.